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PRACTICAL APPLICATIONS IN ASSET AND LIABILITY MANAGEMENT IN INSURANCE COMPANIES

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Abstract

This thesis reviews how an insurance company protects their balance sheet from selected undesired changes. We apply methods to stabilize insurer's own funds under changeable interest rates. The work outlines the principles of interest rates and bonds, which constitute the greatest portion of a Czech insurer's asset portfolio. Literature is reviewed on duration as a measure of its sensitivity to market rate shifts. Commentary on cash flow modeling is given within a brief section on life insurance theory. We examine the sensitivity of classical insurance products, and their portfolio, to parallel and non-parallel yield curve shifts. Duration of modelled liability cash flows is approximated to compare with a government bond portfolio before and after intervention. While Czech insurers' options today in asset and liability management remain limited, we give some methods available to control risk exposure.

Keywords

Actuarial Models, ALM, Bonds, Cash Flow Models, Duration Gap, Interest Rate Risk, Life Insurance

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Introduction

Individuals and companies alike flee to insurers for promises of compensation during undesirable events. While these entities receive protection from risks, the insurer collects the largest base of policyholders possible to run profitable business, because the undesirable events then follow increasingly more predictable patterns. Life insurers, whose products are contracted for decades forward, represent furthermore for the policyholder a savings account, often with a guaranteed yearly appreciation of funds. The insurer invests these inflows in selected assets, making assumptions about their portfolio's future performance, which include, above all, the interest rate.

The 2010s saw minimal volatility in interest rates in Europe. Not long after the economic stress of the COVID-19 pandemic in 2020, central banks raised their target rates to levels not seen since 2008. The insurer's holdings adjust in market value, but not uniformly. The liabilities held by an insurance company are the reserves held in respect of the promises to the policyholders, calculated as the present value of probability-weighted future cash flows, which depend on any issued guarantees and profit participation. These features are absent in typical assets, which are largely bonds in addition to shares, real estate and other items. The difference between the values of assets and liabilities are the insurer's own funds. If the assets diminish more than the liabilities due to the interest rate moving, the insurer risks losing their capital. This thesis aims to use asset-liability management tools to prevent losses in a fictitious insurer's own funds. We model how a Czech insurer's portfolio responds to rate shocks.

We acquaint the reader with relevant interest rate nomenclature and bond theory, including accumulation and its inverse, discounting, in the first chapter. This foundation is useful in the didactic walkthrough of interest rate sensitivity metrics—we attribute each to their author in the second chapter. We distinguish between Macaulay and modified duration, which collapse into one under continuous compounding. We approximate its absolute counterpart, the dollar duration and its scaled analogues such as the 10-point basis point value (“BPV”) and complement with a subsection on convexity. Care must be taken in products with embedded options, where we use effective duration. Finally, we introduce the Fisher-Weil and key rate durations which admit a term structure of interest rates.

In the third chapter, we overview commonplace terminology in life insurance, before commenting on the mechanics of insurance cash flow models, their assumptions, and briefly

their relation to the regulatory framework. Zooming out to practice, we discuss actions the insurer can take to control risk, especially cash flow matching and immunization in the fourth chapter, along with notes on implementation. In the fifth chapter, we use Microsoft Excel to map the effective duration of fictitious bonds against participating policies, risk policies, unit-linked policies, and non-life policies. We then construct their portfolio together with currently tradeable Czech government bonds, which we subject to stress scenarios and compare risk metrics before and after an asset-modifying intervention.

1 Cash Flows, Bonds

1.1 Rate of Interest

The value of CZK 1 at present normally differs from the value of CZK 1 to be received in the future. To give up a monetary unit at present is to abstain from its use or consumption. Intuitively, when money is given elsewhere, *interest* should arise – a reward paid by the *borrower* to the *lender* for the use of capital – an asset, in this case cash. Furthermore, to lend a monetary unit to a different subject might incur the risk they do not pay back, i.e. a risk of default, in which case the lender asks for additional interest on top, called a *risk premium* (Garrett, 2013, p. 1). Without the addition of a risk premium, we speak of *risk-free interest* earned on a borrowed asset, if we know with certainty there will be no default. If we express absolute interest earned over 1 period relative to the borrowed amount, we get the *rate of interest* or *interest rate* i .

We will assume the rate i to be constant for now, implying, as Bierwag (1987, p. 10) calls it, a constant time price for money. If investing one monetary unit for N periods, the interest generated in any non-final period is, applying *compound interest*, assumed to be immediately reinvested at the given rate i for the entire next period. The eventually emergent interest on the reinvestment of the interest is called compound interest. Under this assumption, we relate an asset's value in the present, i.e. *present value*, to its value in the future FV :

$$FV = K(1 + i)^N, \tag{1.1}$$

where N is the number of periods and K is the initial amount.

Say a balance sheet item, i.e. an asset or a liability, is to generate n cash flows in the future. Let C_j be the amount of the j -th flow generated by a balance sheet item at time t_j . Suppose C as the sequence (system) of such cash flows:

$$\begin{aligned} C &= \{ (C_j, t_j), j = 1, \dots, n \} \\ C_j &\in (-\infty, \infty) \\ t_j &\in [0, \infty). \end{aligned} \tag{1.2}$$

Much of literature simplifies the notation by designating time as the index to write C_t . Using the explicit sequence structure $j = 1, \dots, n$, we distinguish cash flows that occur at identical time and irregularly. We say the cash flows are *fixed* if they are deterministic – for example, they cannot be changed by movements in the rate of interest. The word choice is consistent with the widely used term *fixed-income instruments/securities*, which pay predetermined amounts of cash.

Calculating the present value of a project (investment) generally requires the knowledge of:

1. the opportunity cost of the invested capital, which we denote the rate r per *period*
2. the cash flows this project generates over its lifespan

Given n cash flows in total, suppose the present value of the system C as a function of rate r :

$$P(r) = \sum_{j=1}^n \frac{C_j}{(1+r)^{t_j}} . \tag{1.3}$$

The division of the cash flow amount by the term $1+r$ raised to the power of time remaining until cash flow realization is the practice of *discounting*. Discounting enables us to obtain the present value of a future cash flow; it is the inverse operation to accumulation. We may also denote $v = \frac{1}{1+r}$, where we call v the *discount factor*, and then write $P(r) = \sum_{j=1}^n C_j v^{t_j}$.

Assuming risk-free returns and idealizing conditions, such as no transaction costs, we equate $r = i$.

1.2 Continuous Interest

The length of 1 period is commonly set to one year. We may derive rates for other periods and thereby granularize compounding. Suppose m as the number of segments in a year. For example, $m = 4$ splits one year at three points to yield four equally long quarters. Similarly, $m = 365$ would split a non-leap year into its days. We call the interest rate *nominal* and *convertible m -thly* (also called *payable m -thly* or *with m -thly rests*) if the interest accrues at a frequency inconsistent with the period which the rate is defined for, e.g., a bank may charge interest monthly, but the rate is given for one whole year. See an example of semiannual interest in Section 1.5.

If there is, on the other hand, a match between the period and the frequency of compounding, we speak of an *effective interest rate*. Cipra (2010, p. 14) defines the *conformal interest rate* to a given annual effective interest rate i , as

$$i^{(m)} = m \left((1 + i)^{\frac{1}{m}} - 1 \right). \quad (1.4)$$

Consider the following example. If the annual rate is $i = 0.10$, then $i^{(2)} \approx 2 * 0.0488 = 0.0976$. One may think of $i^{(2)}$ as a nominal rate applied twice, in the sense that a half of $i^{(2)}$, here in particular 4.88%, is credited on the initial amount after the first half-year and further 4.88% is credited on the appreciated amount after the second half-year. The result is the same growth as would occur in once-yearly compounding given the annual rate $i = 0.10$. Notice $i^{(2)} < i$, and indeed it holds $i^{(m)} < i$ except for the case of no intra-year splitting, i.e., $m = 1$ where $i^{(1)} = i$.

Cipra (2010, p. 17) lets m grow to infinity to define the force of interest δ corresponding to an effective interest rate i :

$$\begin{aligned} \delta &= \lim_{m \rightarrow \infty} i^{(m)}, \\ i &= e^{\delta} - 1. \end{aligned} \quad (1.5)$$

The limit relationship in (1.4) makes evident why we also refer to δ as a *nominal rate of interest convertible momentarily*. The reader will find more in the work of Garrett (2013, pp. 17-18), who defines the force of interest $\delta(t)$ such that it may vary in time. Garrett defines $i_h(t)$ as a nominal rate of interest per unit time on a transaction of term h (the length of an investment period) beginning at time t . By introducing h , we may allow the rate to vary for different terms and maturities. In our prior discussion, the interest rate varied neither with time nor with the length of investment; here, we transition from the constant i to the time- and term-dependent $i_h(t)$. The *force of interest per unit time at time t* , $\delta(t)$, can then be conceptualized by shortening the investment period infinitesimally:

$$\delta(t) = \lim_{h \rightarrow 0^+} i_h(t). \quad (1.6)$$

Garrett (2013) further defines the accumulation factor $A(t, t + h)$ as the value at $t + h$ (for a positive h) of a monetary unit that was worth 1 at time t . If $\delta(t)$ and $A(t_0, t)$ are continuous functions of t , we can write

$$A(t_1, t_2) = \exp \left[\int_{t_1}^{t_2} \delta(t) dt \right] \quad (1.7)$$

to express the accumulation of one monetary unit from t_1 to t_2 accounting for a continuously varying force of interest $\delta(t)$.

1.3 The Internal Rate of Return (Yield to Maturity)

Consider several future cashflows. Notice that if the first cash flow C_1 occurs at $t_1 = 0$, we may recognize it as the initial investment, i.e. the cash flow at present, oftentimes the initial outflow, in which case $C_1 < 0$. The remaining cash flows occur at future times. The maturity of this cashflow stream equals $t_n - t_1 = t_n$. *Maturity* is the length of time between the purchase of the project and the final cash flow. In this special case, we can separate the first term to define the net present value:

$$NPV(r) = C_1 + \sum_{j=2}^n \frac{C_j}{(1+r)^{t_j}} . \quad (1.8)$$

The *internal rate of return*, also called the *yield to maturity* (YTM), is such a constant rate r that the net present value becomes zero. In the case of a single initial outflow, the outflow is contrasted against the present value of the future incomes to provide a metric that is often used for profitability evaluation. We may interpret YTM calculated at present as a specialized average rate of return on a project (investment), which stays constant over its lifetime, if we buy it now and hold it to maturity (Bodie et al., 2021, p. 438). For conventional cashflows, whose amounts do not change sign across the sequence, only one such rate usually exists, and we will denote it y . There is an intimate relationship between YTM and initial outflow value (e.g. bond price), for which reason bond markets see bonds quoted primarily in their YTM (Martellini et al., 2003, p. 52).

1.4 Bonds

Bonds might be the simplest application of the cash flow formulas. They typically represent the largest portion of the asset portfolio of an insurance company. Bonds are debt instruments giving rise to a relationship between a borrower and a lender. The borrower

issues a bond to satisfy their mostly longer-term capital needs, becoming an *issuer*. We call the bond a *treasury* (also *sovereign bond*, *government bond*) if it is issued by a national government, or a *corporate bond* if issued by a company. The issuer promises the *bond holder* (the lender) a stream of n cash flows at pre-determined future times, the last of which finally contains the *face value* F (also *principal*, *nominal value*). As such, the bond holder trusts to receive future periodic cash payments dependent on the face value F and the *coupon rate* c of the bond in question. The total value of coupons paid out in one year is cF . The *coupons* are the periodic cash payments paid out at the end of each *coupon period*, with one exception; at the specified *maturity date* T , the bond holder receives not only another (the last) periodic coupon as expected, but also the *redemption value* C , completing the series of cash flows, where the redemption value (*redemption price*) is in most standard bonds equal to the face value ($C = F$). Historically, a bond's parameters were specified in a bond indenture where the face value would be printed in very large letters, hence the name (Bierwag, 1987, p. 15). The coupon rate is expressed in decimal form per annum; if the coupon frequency is other than yearly, Bierwag (1987, p. 16) suggests the annual rate be divided accordingly to get the per-period coupon rate. Summarizing in an example, a semiannual bond generates for the bondholder the cashflow $\frac{cF}{2}$ at the end of every coupon period (every 6 months), except at the end of the last coupon period (the maturity date), when the bondholder instead receives $\frac{cF}{2} + F$.

Realizing that the coupon payments resemble a form of interest, let us stress that the coupon rate is neither the internal rate of return nor the prevailing market interest rate, though their relationships influence the bond's properties, including the price. For example, the bond trades *below par value* if the coupon rate is lower than the prevailing market rate and is called a *discount bond*. The opposite is true for a *premium bond* traded *above par value*. If the bond sells at par value (face value), the coupon rate typically equals the yield to maturity. Discount bonds help us conceptualize yield to maturity; the return generated by a bond selling below par consists not only of the coupons, but eventually also of capital gains on the principal – the face value is greater than the initial money invested (Brealey et al., 2020, p. 54). This extends the internal rate of return beyond the coupon rate.

Embedded options allow for flexibility on the lender or borrower side. Callable bonds allow the borrower to pay the debt early, typically with a penalty (call premium). This is a good example of where the redemption price C might differ from (be greater than) the face value F . Redeeming early might be lucrative for the borrower if an opportunity to refinance the debt more cheaply arises, due to their improved credit standing or a fall in interest rates, etc. If the lender holds a *puttable* (retractable) *bond* on the other hand, they have the right to

force the issuer to redeem the bond at par before maturity, which the lender would benefit from if interest rates suddenly rose. After all, a rise in interest rates devalues the bond, but the bondholder, holding a right to sell (put) the bond at a predetermined price, is unaffected and can realize a profit.

1.5 Bond Price, Bond Spread

The market defines the price of the bond accounting for varied factors. For example, a municipally issued bond is expected to be less liquid than a government bond, reducing its price, as would perceived riskiness of the issuing entity. Likewise, countries' issuing governments differ in their creditworthiness among investors. A bond can remain actively traded in a secondary market long after issuance. Its price reflects the right to the remaining future cashflows, recall (1.3): $P(r) = \sum_{j=1}^n \frac{C_j}{(1+r)^{t_j}}$.

For a fixed-income bond specifically, given a semiannual coupon of $\frac{cF}{2}$ currency units and maturity in $\frac{n}{2}$ years, $r_{e,2}$ being the effective semiannual rate, we have:

$$P(r) = \sum_{j=1}^n \frac{\frac{cF}{2}}{(1 + r_{e,2})^{t_j}} + \frac{F}{(1 + r_{e,2})^{t_n}}. \quad (1.9)$$

Alternatively, using $r_{n,2}$ as the annual nominal rate convertible twice yearly, we would write:

$$P(r) = \sum_{j=1}^n \frac{\frac{cF}{2}}{\left(1 + \frac{r_{n,2}}{2}\right)^{t_j}} + \frac{F}{\left(1 + \frac{r_{n,2}}{2}\right)^{t_n}}. \quad (1.10)$$

The above is an example where we may restrict the cash flow amounts to non-negative numbers, so that the cash flows are either all incomes or all outflows. In the discussion on duration in Section 2, we will use

$$C_j \in [0, \infty).$$

Consider a company that is risky to invest in – there is uncertainty as to whether it will fully repay the bond principal and whether it pays coupons out on time. To justify this uncertainty, the company must offer a higher yield to maturity on the bond compared to an otherwise identical risk-free bond. The extra yield is termed *spread* on the security (Ostaszewski, 2002, Section 2) or the *credit spread*. Ostaszewski portrays the credit spread as compensation for the bond issuer's right to default – figuratively, a put option, where the

company's assets can be sold for a predetermined principal despite being of much lower value in case of bankruptcy. The naming "credit spread," however, is misleading as the risk from corporate bonds may come not only from credit risk, but also lower liquidity. The risky bond's yield is, in practice, usually compared to a yield on a government bond, or the *swap rate* – interest rate on the fixed leg of a swap calculated so that the present value of the fixed leg payments equals the present value of expected floating leg payments (Sundaresan, 2009, p. 311).

1.6 Zero-coupon Bond, Spot Rate, Short Rate, Forward Rate

The only cash flow generated by a *zero-coupon bond* is the face value paid out on the maturity date. A zero-coupon bond therefore pays out zero coupons in the interim. We may equate the *n-year spot rate* y_n to the yield to maturity of a zero-coupon bond with a face value of one monetary unit and a maturity date in n years from the present (Garrett, 2013, p. 177).

A memorable difference between a spot rate and a so-called short rate is that the former never pertains to a rate in the future; a spot rate is a rate that prevails today ($t = 0$), i.e., at present, for the term given by the maturity date. In contrast, a *short rate* for a chosen time interval (e.g., 1 year) can come into effect today or instead at some date in 3 years for that fixed time interval of 1 year (Bodie et al., 2021, p. 471). We emphasize that a future short rate does not arise until the future period arrives – up until that time, it is a random variable insofar as we can only guess at or estimate it. This differentiates it from *forward rates* – future reinvestment rates – which are all known, or may at least be determined by inference, at $t = 0$. Forward rates at $t = 0$ are non-random, known rates because they reflect the market consensus (reasoned estimation) at $t = 0$ about what short rates might prevail in the future (Panjer, 1998, p.21).

The forward rates $f_0, \dots, f_{n-1}$ are tied to the n -year spot rates $y_1, \dots, y_n$ by the relationship

$$(1 + y_n)^n = (1 + y_{n-1})^{n-1}(1 + f_{n-1}) = \prod_{j=0}^{n-1} (1 + f_j) \quad (1.11)$$

for which reason we say that forward rates are implied by spot rates. Note that $y_1 = f_0$. Bear in mind that the short rate that will really prevail is not guaranteed to equal the forward rate. By reducing the chosen time interval (which we previously set as 1 year) for the short rate infinitesimally, meaning we consider an infinitesimal maturity, we get the *instantaneous rate of return* (*instantaneous short rate* or simply short rate) at time t , r_t , in the context of

continuous compounding. This is the short rate referred to in continuous-time interest rate models. To clarify, let us again consider a zero-coupon bond with a face value of one monetary unit and a maturity date T , and denote its price at time t by $P(t, T)$. We will assume an interest rate pertaining to the relevant period $R(t, T)$. Using continuous discounting, the price of the zero-coupon bond is

$$P(t, T) = e^{-R(t, T)(T-t)}, \quad t < T. \quad (1.12)$$

Take the natural logarithm of both sides to get

$$R(t, T) = -\frac{\log P(t, T)}{T-t}. \quad (1.13)$$

The instantaneous rate of return at time t can be obtained by bringing T arbitrarily close to t in (1.13), shrinking time to maturity to zero:

$$r_t = \lim_{T \rightarrow t^+} R(t, T) = \lim_{T \rightarrow t^+} -\frac{\log P(t, T)}{T-t}. \quad (1.14)$$

Factor out the negative sign and recognize that $P(t, t) = 1$ and therefore $\log P(t, t) = 0$ to write

$$r_t = -\lim_{T \rightarrow t^+} \frac{\log P(t, T) - \log P(t, t)}{T-t}$$

where the limit corresponds to the definition of the derivative of $\log P(t, T)$ with respect to T evaluated at $T = t$ and hence

$$r_t = -\frac{\partial}{\partial T} \log P(t, T) \Big|_{T=t}. \quad (1.15)$$

This is in fact an application of the *instantaneous forward rate* at time t defined by Heath et al. (1992, p. 79):

$$f(t, T) = -\frac{\partial}{\partial T} \log P(t, T).$$

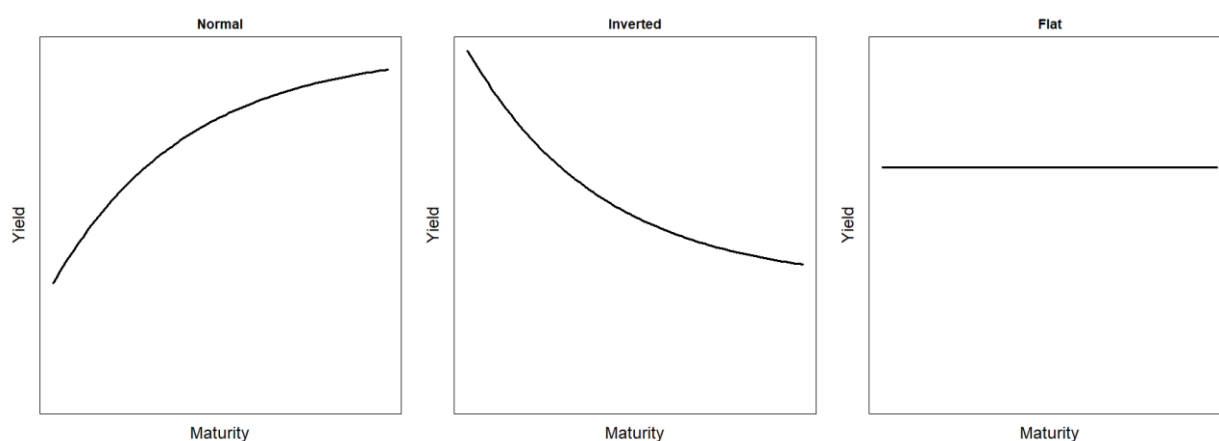
For further reading, refer to Brigo and Mercurio (2006), Chapter 1, and Hull (2022), Chapter 4.

1.7 Term Structure of Interest Rates

Under typical market conditions, one might intuitively demand a higher rate of return on their investment if they commit their money for longer – in other words, the time to maturity should influence the interest rate. A plot of the *term structure of interest rates* places the interest rate for a given security on the vertical axis and time to maturity on the horizontal axis. For bonds, the yield to maturity is used on the vertical axis of the term structure plot to give rise to a *yield curve* at a particular time (Cipra, 2010, p. 61). In practice, these may change day-to-day in response to economic conditions. If we plot either the yields to maturity or the spot rates at some time $t = 0$ against maturities, we will roughly get either of the three shapes in Figure 1.1.

Figure 1.1

The normal, inverted and flat yield curve



Note. Adapted from Fabozzi & Fabozzi (2023, p. 78).

1.8 Stocks, Value-at-Risk

By holding a company's stock (share), one holds partial ownership of the company as well as the right to participate in the company's management decisions. Stocks commonly form part of an insurer's portfolio. We briefly present Value-at-Risk.

Take an insurer's stock portfolio at $t = 0$ and let the composition remain frozen onward, with evolving portfolio value. In one year, that is, after 252 trading days at $t = 252$, the portfolio

value is a random variable W with the probability density function $f_W(w)$. Consider the expression

$$1 - \alpha = \int_{w_\alpha}^{\infty} f_W(w) dw , \quad (1.16)$$

which implies that $1 - \alpha$ is the probability that the outcome of W falls in the interval (w_α, ∞) , i.e., $\Pr[W > w_\alpha] = 1 - \alpha$. We call $1 - \alpha$ the confidence level; w_α is a quantile. Jorion (2007, p. 108) calls the following expression the relative Value-at-Risk:

$$VaR_{rel,\alpha} = E[W] - w_\alpha , \quad (1.17)$$

reserving the term absolute Value-at-Risk for the expression:

$$VaR_{abs,\alpha} = W_0 - w_\alpha , \quad (1.18)$$

where W_0 is the portfolio value at $t = 0$.

The factor α determines the magnitude of the considered loss. For example, if $\alpha = 0.005$, the probability that the final portfolio value W is lower than w_α is 0.005. The corresponding confidence level is 99.5% – this level is required by Solvency II for Solvency Capital Requirement calculation. Value-at-Risk can be interpreted as the worst loss over – in our case – a year, such that the probability that the actual loss will be greater is only α . It is the loss that is expected to occur no more than $\alpha \cdot 100\%$ of the time over our horizon. For a horizon of a day, if a daily VaR is €100 000 at $\alpha = 0.05$, the probability that the loss the next day will be greater than €100 000 is only 0.05. Value-at-Risk is an estimator constructed from historically observed data. The simplest models assume that asset returns follow a normal distribution. See more in the work of Jorion (2007).

2 Duration

2.1 Macaulay Duration

Frederick R. Macaulay (1938, p. 48) used the yield to maturity to first define loan duration. The *Macaulay duration* of the system of fixed cash flows C may be interpreted as a weighted average of times to each cash payment. On pages 46–47, Macaulay justifies the use of present (rather than future) value for weighing; the weight is the present value of a given cash payment relative to the total present value of the cash flow system. Let $P(y)$ be the present value of the cash flow system C using the yield to maturity y adjusted per (coupon) period for discounting. With the restriction $C_j \in [0, \infty)$ in place, we can recall the definitions of the Macaulay duration:

$$D(y) = \sum_j t_j \frac{C_j(1+y)^{-t_j}}{P(y)} = \frac{\sum_j t_j C_j(1+y)^{-t_j}}{P(y)} = \frac{\sum_j t_j C_j(1+y)^{-t_j}}{\sum_j C_j(1+y)^{-t_j}}. \quad (2.1)$$

Macaulay duration is independent of monetary units used, as its units are instead determined by the period selected for calculation. For bonds paying semiannual coupons, we use one-half the annual yield to maturity for y and the units of duration will then be half-years. As note Fabozzi and Fabozzi (2021, p. 122) in Chapter 4:

$$\text{Duration in years} = \frac{\text{Duration in } m \text{ periods per year}}{m}.$$

We can differentiate the sum $P(y) = \sum_j C_j(1+y)^{-t_j}$ with respect to y to obtain

$$\frac{dP(y)}{dy} = - \sum_j t_j C_j(1+y)^{-t_j-1} = -(1+y)^{-1} \sum_j t_j C_j(1+y)^{-t_j}. \quad (2.2)$$

The factor $\sum_j t_j C_j(1+y)^{-t_j}$ is also present in the numerator of (2.1). After algebraic manipulation of (2.1) and (2.2), we see the motivation behind why duration is more generally defined as:

$$D(y) = - \frac{dP(y)}{dy} \frac{1+y}{P(y)}. \quad (2.3)$$

We may view duration as the average or effective maturity of a balance sheet item such as a bond. As such, we sometimes refer to duration as the *discounted mean term* (of a project). Having a singular cash flow at the maturity date, a zero-coupon bond has its Macaulay duration equivalent to the maturity date.

To further interpret Macaulay duration, suppose that an investor just now obtained a bond. A sudden notification of an interest rate rise would be negative news, as it would devalue the bond in the short-term (due to the change in discount factor), but not in the long run, thanks to coupon reinvestment at the higher rate and an eventual return of the bond price to par (La Grandville, 2001, p. 143). Duration determines the horizon at which the investor becomes indifferent to the rate increase as the coupon reinvestment gains completely offset the bond price fall. Notice that $\frac{dP(y)}{dy} < 0$, which corresponds to the intuition that rising rates should devalue the bond and vice versa.

2.2 Duration as a Function of Market Rate, Modified Duration

Let us clarify the appearance of duration in the analysis of interest rate sensitivity of a cash flow system. Recall the rate r per period as the opportunity cost of capital. For straightforward illustration, make the simplifying assumption that the opportunity cost of capital is equal to a prevailing market rate r . We collapse the term structure such that the rate r is constant and omit the ideas of risk premiums and embedded options. The yield to maturity on the instrument should then coincide with the constant market rate r , allowing us to connect duration to the exogenous market environment and treat duration as a function of r :

$$D(r) = -\frac{dP(r)}{dr} \frac{1+r}{P(r)}. \quad (2.4)$$

Suppose now a change to the present value of the cash flow system in response to a movement of the market rate by Δr . Let us express the present value as a function of the rate r shifted by Δr using the Taylor expansion around the point r :

$$\begin{aligned} P(r + \Delta r) &= P(r) + \frac{dP(r)}{dr} (r + \Delta r - r) + \frac{1}{2!} \frac{d^2 P(r)}{dr^2} (r + \Delta r - r)^2 + \dots \\ P(r + \Delta r) &= P(r) + \frac{dP(r)}{dr} \Delta r + \frac{1}{2!} \frac{d^2 P(r)}{dr^2} (\Delta r)^2 + \dots \end{aligned} \quad (2.5)$$

Subtracting $P(r)$ from both sides, approximate now the change of the present value in response to a market rate movement, omitting terms of order higher than 1:

$$\Delta P(r) \approx \frac{dP(r)}{dr} \Delta r .$$

Recall (2.4); rearranging and substituting for $\frac{dP(r)}{dr}$, we express this first term of the Taylor expansion as:

$$\Delta P(r) \approx \frac{dP(r)}{dr} \Delta r = -P(r) \frac{D(r)}{1+r} \Delta r = -P(r) D_{mod}(r) \Delta r$$

where we have just let $D_{mod}(r) = \frac{D(r)}{1+r}$ be the *modified duration*:

$$D_{mod}(r) = \frac{D(r)}{1+r} = -\frac{dP(r)}{dr} \frac{1}{P(r)} = -\frac{d}{dr} [\log P(r)] .$$

(2.6)

We can then express the approximate absolute and percentage changes in an instrument's price as follows:

$$\Delta P(r) \approx -D_{mod}(r) P(r) \Delta r ,$$

$$\frac{\Delta P(r)}{P(r)} \approx -D_{mod}(r) \Delta r .$$

(2.7)

If we let $\Delta r = 0.01$ in (2.7), we see that modified duration estimates the percentage change of the present value in response to a change of the interest rate by 1 percentage point (a change of 100 basis points). We consider it a simple measure of interest rate sensitivity. For example, between the coupon yields 2% and 8% of two bonds with identical maturity dates, the 2% bond will have higher modified duration and correspondingly higher sensitivity to interest rate fluctuations. This is due to a greater portion of the inflows being placed at the distant maturity rather than in the nearer yearly coupon payments. For this reason, the 2% bond would appreciate more in the case of sudden rate decline. We assume here that the cash flows themselves do not change due to the interest rate shift, as might occur in bonds with embedded options.

The expression on the very right of (2.6) implies that modified duration can be equated to the opposite of the slope of the tangent line to the curve of the graph $\log P(r)$, as it is expressed by Ostaszewski (2002, Part 2). Though it is modified duration that measures price sensitivity of the bond to rate shifts, one often sees a blurry distinction between Macaulay and modified duration in literature. One reason is that these converge under continuous

compounding (Kakushadze & Serur, 2018). Indeed, Fabozzi and Konishi (1996, p. 35) refer to the difference between them as often inconsequential.

Recall the term structure of interest rates defined in Section 1.7. Our reasoning so far maintained that the entire term structure plot shifts by an infinitesimal movement Δr (Veronesi, 2016, p. 121). We will address the arising deficiencies in Section 2.3. Moreover, as we have assumed the rate r to be constant, our term structure is currently flat – a straight line. As the term Δr is added uniformly to a constant rate r in (2.5), the yield curve shifts in a parallel manner, but we will relax this in Section 2.7.

2.3 Convexity

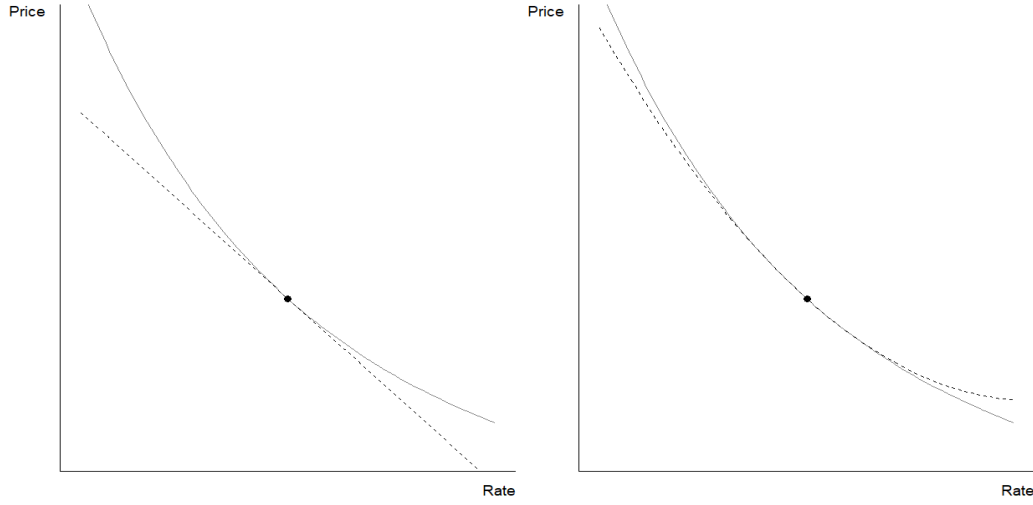
Let us revisit the Taylor expansion visually. The second derivative of (1.3), $\frac{d^2 P(r)}{dr^2}$ is positive (for $r > 0$, $C_j > 0$) giving the price function $P(r)$ a convex character at all points r of reasonable interest. The bond price as a function of interest rate r is indeed a convex function. Larger interest rate movements make the linear treatment increasingly imprecise. Including the second-order term of the Taylor expansion, we have:

$$\Delta P(r) \approx \frac{dP(r)}{dr} \Delta r + \frac{1}{2} \frac{d^2 P(r)}{dr^2} (\Delta r)^2 . \quad (2.8)$$

See Figure 2.1 for a comparison between the use of one and two orders of the Taylor expansion to estimate the bond price as a function of rate r used for discounting.

Figure 2.1

Comparison of Taylor series approximations of $P(r + \Delta r)$ about the point r



Note. Own work using RStudio and the package ggplot2 (Wickham, 2016). The approximations are represented by dashed lines. $F = 1000$, $c = 0.05$, $T = 10$ years. We omit orders higher than 1 on the left, while omitting orders higher than 2 on the right.

The consequence of convexity is that the bond price becomes more sensitive as the rate falls, and equally, the price sensitivity decreases as the rate rises. Fabozzi and Konishi (1996) note this is beneficial to an asset holder. Specifically, say that market rates increase – the asset value declines, but convexity dampens the holder’s loss.

Introducing the measure of *convexity*

$$CV(r) = (1 + r)^{-2} \frac{\sum_j t_j(t_j + 1)C_j(1 + r)^{-t_j}}{P(r)} = \frac{d^2 P(r)}{dr^2} \frac{1}{P(r)}, \quad (2.9)$$

we can multiply both sides of (2.9) by $P(r)$ and substitute into the second term of the abovementioned Taylor expansion in (2.8), so we can improve the approximation of the price change:

$$\Delta P(r) \approx -P(r)D_{mod}(r)\Delta r + \frac{1}{2}CV(r)P(r)(\Delta r)^2. \quad (2.10)$$

Some authors, such as Cipra (2010, p. 33), refer to convexity as we defined it in (2.9) as the *modified convexity*, being consistent with the duration nomenclature. After all, we use it side-by-side with modified duration for the same purpose in the Taylor expansion in (2.10). Adding convexity allows us to approximate the change in present value with greater

precision for larger interest rate changes. The approximate relative price change due to convexity equals:

$$\frac{\Delta P(r)}{P(r)} = \frac{1}{2} CV(r) (\Delta r)^2 . \quad (2.11)$$

If given in percent, (2.11) may also be referred to as *percentage price change not explained by duration*. Of course, nothing stops us from increasing the accuracy of approximation by calculating derivatives of order higher than 2, but this is rarely done in industry as two derivatives are considered sufficient for most yield movements (Fabozzi & Konishi, 1996, p. 87).

Fabozzi and Fabozzi (2023, p. 207) approximate bond convexity as follows:

$$CV(r) \approx \frac{P(r + \Delta r) + P(r - \Delta r) - 2P(r)}{P(r) (\Delta r)^2} . \quad (2.12)$$

2.4 Dollar Duration

Where Macaulay and modified durations are relative metrics, *dollar duration* is a commonly used absolute measure that reflects the actual change in monetary terms. In dollar duration, we weigh modified duration by the holding's value:

$$D_{\$}(r) = D_{mod}(r)P(r) = -\frac{dP(r)}{dr} . \quad (2.13)$$

We may interpret dollar duration as the change in price of a holding following a change in the rate r of one unit. Of course, such rate movements are unrealistic if r is expressed in decimal form, as it normally is. And so, dollar duration is often scaled for easier interpretability, often by multiplying by 0.01. Some authors prefer this meaning of dollar duration, e.g. Fabozzi and Konishi (1996, p. 37). The resulting value then captures the change in price of a holding if the rate moves by one percentage point. If we instead opt for the factor 0.0001, we can speak of the *basis point value (price value of a basis point)*, often abbreviated BPV (PVBP) or referred to as DV01, i.e., dollar value per 01 (1 basis point). This would be consistent with the notation of Tuckman and Serrat (2012, p. 126), who write:

$$DV01 = -\frac{1}{10000} \frac{dP(r)}{dr} . \quad (2.14)$$

As the name suggests, DVo1 (BPV) gives the change in price of a holding given a rate change of one basis point, i.e., 0.01 of a percentage point. However, many use the BPV as a loose label rather than a standardized metric. To ease interpretation, there is definitional flexibility in industry use in the scaling factor, terminology and the sign convention. In our case, notice again the negative sign in $-\frac{dP(r)}{dr}$ which ensures that $D_{\$}$ is a positive number under the circumstances that prices increase when rates decline, which holds for all fixed coupon bonds.

Because the price change is almost symmetrical for small yield changes, we can approximate dollar duration as an average of the changes resulting from the yield moving up and moving down:

$$D_{\$} \approx -\frac{P(r + \Delta r) - P(r - \Delta r)}{2\Delta r}. \quad (2.15)$$

Dollar duration adds up across holdings. The additive nature is useful; it implies that the dollar duration of net worth of a company equals the dollar duration of assets minus the dollar duration of liabilities.

2.5 Duration Patterns

Note that *ceteris paribus*:

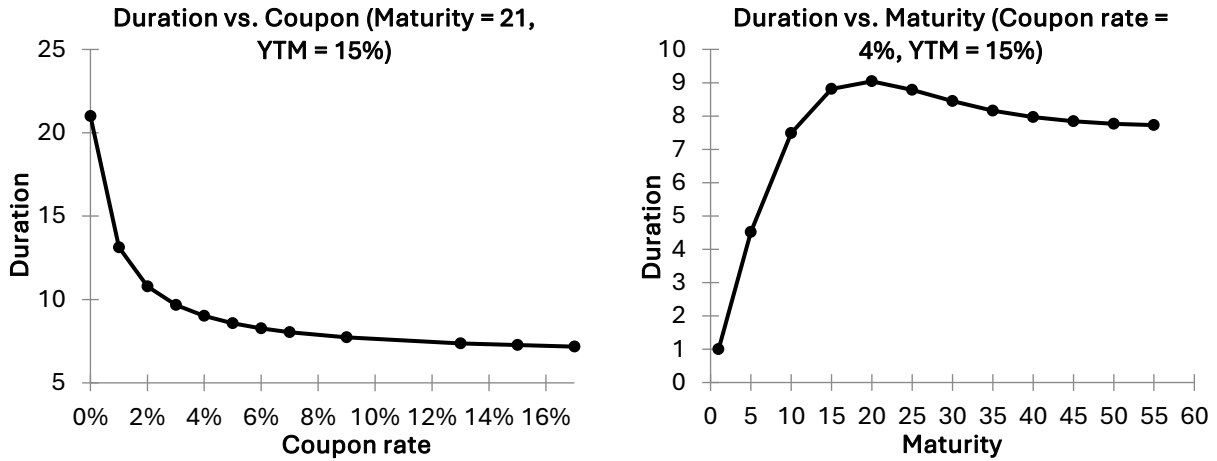
1. The greater the bond's yield to maturity, the lower the (Macaulay, modified, dollar) duration.
2. The greater the time to maturity, the greater the (Macaulay, modified, dollar) duration. However, for bonds with the coupon rate less than the yield to maturity, this ceases to apply for maturities over, typically, 25 years, see more in the work of La Grandville (2001, pp. 83-86) and Hawawini (1981).
3. The greater the coupon rate, the lower the (Macaulay, modified) duration.

Relationship 3 above is reversed for dollar duration, presumably due to a rising coupon implying a rising price under fixed yield to maturity. The increase in price P in the product $D_{mod}P = D_{\$}$ then outweighs the decline in D_{mod} (which occurred due to coupon rate increase), ultimately increasing $D_{\$}$. The reader is encouraged to see further demonstration in Martellini et al. (2003, p. 172) but exercise caution that these authors opt to define dollar duration without including the negative sign in $-\frac{dP(r)}{dr}$, and hence consider dollar duration

to be a negative number in most intuitive cases such as fixed coupon bonds. Some of the patterns are illustrated in Figure 2.2.

Figure 2.2

The effect of coupon rate and maturity on duration



Note. Adapted from Benninga (2008, p. 678).

2.6 Effective Duration

The term *effective duration* (*option-adjusted duration*) was devised to account for embedded options in an instrument. Here, the expected cash flows are allowed to change in response to interest rate movements. For option-free bonds, the effective and modified duration are equivalent; the former is a generalized term. In practice, effective duration is approximated as follows (Gajek et al., 2005):

$$D_E \approx -\frac{P(r + \Delta r) - P(r - \Delta r)}{2\Delta r P(r)}. \quad (2.16)$$

Even though Macaulay duration may be conceptualized as the weighted average of payout timings, applying such interpretation to effective duration fails when we realize it is possible for a bond to have its effective duration exceed its maturity (Fabozzi & Fabozzi, 2021, p. 146). It becomes best to think of duration as a measure of sensitivity to interest rate movement.

2.7 Fisher-Weil Duration, Key Rate Durations

Until now, we have assumed the prevailing market rate r to remain constant across time to use the same discounting rate for each cash flow. Let us relax this simplification to allow for a non-flat term structure of interest rates. We use the term *Fisher-Weil duration* when we replace the yield to maturity in the Macaulay duration calculation with spot rates implied by

the yield curve for each cash flow (Panjer, 1998, p. 114). We can make use of the notation $P(t, T)$ introduced in Section 1.6 in defining the *Fisher-Weil duration*:

$$D_{FW} = \frac{\sum_j t_j C_j P(0, t_j)}{\sum_j C_j P(0, t_j)} . \quad (2.17)$$

Panjer also recalls the *Fisher-Weil dollar duration*:

$$D_{\$, FW} = \sum_j t_j C_j P(0, t_j) . \quad (2.18)$$

But rather than thinking of interest rate sensitivity as a single number, one might wish to represent it as a vector of numbers. Ho (1992) introduced the measure of *key rate durations* (also *partial durations*), which have since become used increasingly in practice. Where effective duration is the total sensitivity of a bond, key rate durations are components of this total. Key rate durations correspond to i specific points – *key rates* $t(i)$ – along the yield curve, and Ho denotes each shift of a key rate by $S[t(i)]$. Then, the resulting overall continuous yield curve shift along the entire maturity range can be approximated by interpolating linearly between all the shifts. Given the price P , the price sensitivity to an infinitesimal change in any of the i key rates is the *key rate duration* $KRD(i)$:

$$\frac{\Delta P(i)}{P} \approx -KRD(i) d(i) , \quad (2.19)$$

where $d(i)$ is the magnitude of the shift at the given key rate. Key rate durations deconstruct the total (interest) risk exposure and aggregate linearly; the sum of key rate durations equals the effective duration. Equivalently, the overall relative change in price is the sum of all *key rate changes* $KRD(i)d(i)$:

$$\frac{\Delta P}{P} \approx \sum_i -KRD(i) d(i) . \quad (2.20)$$

3 Life Insurance

3.1 Principles of Life Insurance

The business of life insurance handles risk exposure for profit. The business hands its client a promise of *benefit* payment in case of a certain event in exchange for a *premium*. In life insurance, the supposed event, termed the *insured event*, is most often the death or survival to a pre-specified age of the *insured person*. The client herein is termed the *policyholder* but is naturally not always the same as the insured person, nor the *beneficiary*, i.e., the person who has the right to the stated benefit payment from the *insurer* in case the insured event occurs. Let us refer to the promise binding the policyholder with the insurer as the *policy*. Committing the insurer to a future payment, whose timing and occurrence itself are uncertain, the policy constitutes a single liability.

Suppose such a policy is to be established. Abstract away from expenses arising from the policy creation and upkeep. To ensure that the expected future loss of the insurer is zero, the insurer uses the *equivalence principle* to calculate the *net premium*, where all expenses other than the benefit payments are ignored. This means that the present value of the expected future benefits to be paid (*outflows, outgo*) equals the present value of the expected future premiums to be received (*inflows*).

We collectively refer to outflows and inflows as *cash flows*. If the insured event does not occur as time passes, the present value of the benefits to be paid rises as we approach the payout time. At the same time, typically fewer and fewer premiums remain to be collected. This development illustrates the need for the insurer to continuously monitor the difference between the present value of inflows and present value of outflows which is represented by any of the terms *policy value*, *prospective reserve* and *technical reserve* (Neill, 1992, p. 113).

To quantify the uncertainty in the calculation of the expected future cash flows, we consider the development in time of factors such as mortality rates, interest rates, policyholder behavior, etc. If we speak solely of the uncertain development of human lives, we refer to the events herein as *life contingencies*. In general, the randomness is twofold; there is variation about the mean rate of the contingency, and there may be shifts of the mean rate itself (Redington, 1952). *Life tables* typically determine the best estimated mortality rates to be used in the valuation of life liabilities.

The law of large numbers enables the insurer to confide in future expected values. As the number of policies issued grows, so does the probability that the actual cash flow movements per policyholder match the expected cash flow movements per policyholder, which are used to calculate the net premium (Tinungki, 2018). The great number of policyholders collectively share the burden of risk emerging from insured events, which Briys and De Varenne (1997) refer to as *risk mutualization*. Dickson et al. (2009, pp. 334–336) use the central limit theorem to demonstrate mortality as increasingly deterministic with a growing population of lives to treat most mortality risk as *diversifiable*. On the other hand, we treat interest rate risk as non-diversifiable, as it cannot be eliminated by increasing the number of policies. This contributes to interest rate risk as the most prominent aspect of ALM in insurance.

3.2 Cash Flow Projection Models

Cash flow projection models calculate for each of the discrete time instants t , which are spaced out by a time interval of a month or a year, the difference between the expected future inflows and expected future outflows. The reader may refer to Teichmannová (2021) and Hoang (2020) who describe cash flow projection models in their works and Janeček (2006) and Cipra (2006) for their broader modifications. We will refer to the parameters of such a model as *assumptions*. For example, we do not know what the interest gained on money will be in the future, nor do we know what mortality a group of people will experience, and hence we need to assume the future development. See some such parameters below.

3.2.1 Assumption: Mortality rates

It is in life tables we find the probability of death q_x (*mortality rate*) of an x -aged individual. Consider $x \in 0, 1, 2, \dots$, then q_x is the probability that a person of the precise age x , denoted (x) , does not live to the precise age of $x + 1$, meaning they die before reaching that age. Since dying and surviving to age $x + 1$ are mutually exclusive events, it follows that the probability that (x) survives at least to age $x + 1$ is $p_x = 1 - q_x$. Suppose the interval h , then the probability that (x) survives at least to age $x + h$ is denoted ${}_h p_x$, the complementary event of probability ${}_h q_x$ being that (x) dies within h years. Note ${}_1 p_x \equiv p_x$ and ${}_1 q_x \equiv q_x$. See Neill (1992) for further reading. Gerber (1997, p. 21) presents three methods of extracting monthly probabilities of death $q_x^{(m)}$ from the annual ones given in life tables. The simplest is the uniform distribution method, where we set $h = \frac{1}{12}$ and assume $q_x^{(m)} = {}_{1/12} q_x = \frac{1}{12} q_x$ for $m \in 1, 2, \dots, 12$.

Bowers et al. (1997, p. 52) use the following definition, which is common in actuarial texts: let X be a continuous random variable representing a newborn's age at death. Suppose $x \in [0, \infty)$ – the future lifetime of an x -aged-life (x) is then the continuous random variable $T(x) = X - x$. Then, ${}_h p_x = \Pr[T(x) > h]$ and ${}_h q_x = \Pr[T(x) \leq h]$ where $h \geq 0$.

Mortality rates typically overestimate the actual rate of dying of insured persons in a particular life insurance business, the result of selection bias possibly due to wealth and the exclusion done by health underwriting.

3.2.2 Assumption: Lapses

At any time, the policyholder is permitted to lapse their policy. In such a case, the insurer pays out a *surrender value*, which typically corresponds to the accumulated reserve. A *lapse rate* in the period t after policy inception, denoted s_t , is the probability that this client lapses their policy in the given time interval. This includes both cases where a client cancels the policy on purpose and cases where it ends automatically, for example if premiums are not paid. If lapses occur in high numbers or unexpectedly, the risk of liquidity stress is introduced, further adding to the importance of technical reserve monitoring. Lapse rates are often estimated from the life insurer's prior observation.

3.2.3 Assumption: Commissions

Commissions are outflows paid out to intermediaries (agents). There are two types: acquisition commissions and maintenance (renewal) commissions. Some of the commission may have to be returned to the insurer if a lapse occurs – we term such an inflow a commission claw back. Insurance companies account for the possibility the claw back might never materialize, for instance if the agent declares bankruptcy (Janeček, 2006, p. 22).

3.2.4 Assumption: Expenses and Inflation

Dickson et al. (2009, p. 150) divide expenses as follows:

1. Initial expenses: underwriting expenses (medical tests), may also include commissions to agents for selling the policy as already noted above, ...
2. Renewal expenses: staff salaries, rental costs and upkeep of premises, processing costs connected to annuity payments and annual statements to policyholders, ...
3. Termination expenses: small costs associated with paperwork at policy expiry.

The future development of the inflation rate is itself an assumption and bears significance in expenses, influencing, say, costs of labor.

3.2.5 Assumption: Interest rates

Fluctuations in interest rates have far-reaching implications for the insurance business. The assumed yield curve determines the rate used for discounting cash flows, which determines the value of the liability. It also determines the calculated yield on assets. The simplest models assume a predetermined term structure and its variations in scenario testing; simple stochastic models, such as the Vasicek and Ho-Lee, simulate the development of the instantaneous short rate as a random variable.

3.2.6 Commutation Functions

Commutation functions involve both the interest and mortality functions and ease computation of insurance premiums. Consider a constant interest rate i , let $v = \frac{1}{1+i}$, let d_x be the number of deaths between ages x and $x + 1$, and l_x the remaining survivors. The *radix*, i.e. the entering survivors l_0 , may be chosen arbitrarily. The 0th-level commutation functions are defined as:

$$D_x = l_x v^x , \tag{3.1}$$

$$C_x = d_x v^{x+1} . \tag{3.2}$$

The 1st-level commutation functions are defined as follows:

$$N_x = \sum_{k=0}^{\infty} D_{x+k} , \tag{3.3}$$

$$M_x = \sum_{k=0}^{\infty} C_{x+k} . \tag{3.4}$$

In tabular software such as Microsoft Excel, we may use these functions for net premium P_{net} calculation when applying the equivalence principle under zero lapses and first-order assumptions (see Section 3.5):

$$P_{net} = SA \cdot \frac{M_x - M_{x+n} + D_{x+n}}{N_x - N_{x+n}}, \quad (3.5)$$

for *endowment* insurance i.e., the insured receives a benefit in the case of death and in the case of survival to a specified age, n being the policy period, SA the sum assured. If only the risk of death is covered, we speak of the *term* insurance and use:

$$P_{net} = SA \cdot \frac{M_x - M_{x+n}}{N_x - N_{x+n}}. \quad (3.6)$$

For further reading, see Neill (1992, Chapter 2).

3.2.7 Cash Flow Models

Consider a single life insurance policy of the endowment type and assume yearly time intervals. We gather the policy parameters together in one place:

1. Inflow: Regular yearly premium
2. Outflow: Commissions expenses
3. Outflow: Expenses
4. Outflow: Death benefit
5. Outflow: Surrender value
6. Single outflow: Survival (maturity) benefit

The amounts of inflows and outflows are contingent on events whose probabilities are determined by our assumed mortality rates and lapse rates from assumptions. We ensure that the mortality rates and lapse rates are defined with respect to the selected interval of one year. Given 100% of hypothetical policies at the beginning of the projection period at time $t = 0$, i.e. $N_0 = 1$, for the next year, we decrement a portion due to mortality and another portion due to lapses, which decreases the relative number of hypothetical policies in force the next year. If N_t is the relative number of policies in force at the start of the year t , we can write $N_t < 1, t \neq 0$. This is done iteratively year by year until the observed policy reaches its maturity.

Consider the lapse timing parameter, where $\theta = 0$ means lapses occur at the beginning of the year, whereas $\theta = 1$ means lapses occur at the end, after that year's deaths, which we assume in Chapter 5. Given mortality rates q_x , lapse rates s_t , and the parameter $\theta \in [0,1]$:

1. we model the yearly death decrement for (x) as

$$D_t = N_t q_x (1 - (1 - \theta) s_t), \quad (3.7)$$

2. and the yearly lapse decrement at t as

$$L_t = N_t s_t (1 - \theta q_x). \quad (3.8)$$

In some products, such as participating policies, the death benefit, the survival benefit and the surrender value include an accumulating fund value in addition to a fixed sum assured. At the beginning of each year t , the policyholder contributes to the fund value the premium. A risk charge is deducted from the fund – in the model we use in Chapter 5, we weigh the sum assured by the probability of its payout. The insurer also deducts a management fee. The fund value then appreciates at a minimum guaranteed rate, and, if applicable, by a proportion – the *profit share ratio* π – of the investment return generated beyond the guaranteed rate. To reiterate, consider the post-charge fund value PC_t :

$$PC_t = Fund_{t-1} + P - f \cdot Fund_{t-1} - q_{x+t-1} \cdot SA; \quad (3.9)$$

we calculate the policyholder's fund $Fund_t$ at the end of the year t as

$$Fund_t = PC_t (1 + y_{gar} + \pi \cdot \max\{y_{ret} - y_{gar}, 0\}), \quad (3.10)$$

where

- P is the yearly premium,
- f is the yearly management fee rate,
- y_{gar} is the guaranteed yearly return,
- y_{ret} is the assumed yearly investment return,
- π is the profit share ratio.

To obtain the cash flows themselves, we use the assumptions and the policy parameters. We compute the expected cash flows for each year:

1. Inflow: multiply the premium by N_t ,
2. Outflow: multiply the commissions expenses by N_t ,
3. Outflow: multiply expenses by N_t ,
4. Outflow: multiply the death benefit by D_t (the death decrement of that year),
5. Outflow: multiply the surrender value by L_t (the lapse decrement of that year),
6. Single outflow: multiply the survival benefit by N_T (the relative number of policies at policy maturity).

The cash flow at time t , CF_t , is the difference between the inflows and the outflows. As for time value, we follow market practice to assume that the first three cash flows in the above list occur at the beginning of the year and accrue interest, while the remaining three occur at the end. The liability value that this single policy represents – policy value, as we further discuss it in Section 3.3 – is obtained by discounting all cash flows to the present. If rather than a single policy, we consider j policies, we get the present value of future cash flows of the j -th policy:

$$PVFC_j = \sum_{t=1}^T \frac{CF_{j,t}}{(1 + i_t)^t}, \quad (3.11)$$

where i_t is the spot rate for maturity t under second-order assumptions (see Section 3.5). Summing across all policies gives the total value of liabilities of the insurance company.

3.3 Policy Value

Dickson et al. (2009, p. 145) define the future net loss random variable at policy inception ($t = 0$), PV being the present value:

$$L_0^n = PV \text{ of benefit outgo} - PV \text{ of net premium income}.$$

The future gross loss accounts for expenses:

$$L_0^g = PV \text{ of benefit outgo} + PV \text{ of expenses} - PV \text{ of gross premium income}.$$

All the summands of L_0^g are random variables. That is easy to see given how long the policyholder survives is unknown at policy inception, as are all the assumptions. The equivalence principle, first mentioned in 3.1, sets $\mathbb{E}[L_0^g] = 0$ to obtain the premium to be requested. But the expected future loss being zero is only true for the instant of policy

inception. Beyond policy inception ($t > 0$), assuming that the policy is still in force, $\mathbb{E}[L_t^g]$ is normally no longer zero due to the new information at time $t > 0$ – namely, the insured is alive at that specific instant, and fewer premiums remain. The insurer's expectations of *PV of benefit outgo* and *PV of net premium income* have changed. It is normally the case that if a policy is still in force t years after being issued, future premiums (from time t) cannot fully cover future benefits and expenses (Dickson et al., 2009, p. 179). The shortfall is the *policy value* at time t , denoted ${}_tV$:

$${}_tV = \mathbb{E}[PV \text{ of benefit outgo}] + \mathbb{E}[PV \text{ of expenses}] - \mathbb{E}[PV \text{ of gross premium income}]. \quad (3.12)$$

The terms *technical reserve* and *prospective reserve* may be used interchangeably with policy value ${}_tV$ or may be distinguished to mean the actual capital held for this purpose, as it may be lower or higher than ${}_tV$. To reiterate, reserves are held and accumulated from the beginning of the policy's lifetime to meet liabilities in the later stages of policy's life, when outgo on benefits typically exceeds income from premiums. The insurer's business is financially sound if their investments (assets' side) are greater than the total of their policy values (liabilities' side). The comparison is done during *company valuation*. The amount by which the assets exceed the liabilities is the *surplus, equity, net worth*, or the *own funds* – we use these synonymously.

3.4 Embedded Options in Life Insurance

When the benefits under an insurance policy contain a financial guarantee, what arises is an *embedded option*. Interest rate guarantees and profit sharing are common option-like elements in Czech insurance policies. We use the terms *with-profit life insurance*, *profit-share policies* and *participating policies*. Incorporating these features in an insurance contract understandably changes the value of the emerging liability, primarily through the *time value of financial options and guarantees* (TVFOG, TVOG).

3.4.1 Profit sharing

Profit sharing (profit share) enables policyholders to participate in the investment performance of the insurance company. It is common for an insurance company to promise a percentage of the profit it makes above the guaranteed *technical interest rate*. The insurer may first deduct a fee for fund management before adjusting by the profit share ratio to get the amount credited to the policyholder's fund. Švec (2011, p. 7) states the profit share ratio to often hover between 70% and 90%.

The feature of profit sharing can decrease the sensitivity of the liability to a rise in interest rates, since the discounting done at a higher rate, which lowers the liability value, is offset by the profit share guarantee increasing the liability value. This is especially pronounced if the profit share is heavily formulaic and there is low discretion in how much profit is to be shared.

3.4.2 Interest Rate Guarantees

An interest rate guarantee ensures that a policyholder receives at least a specified minimum return on their investment per period, regardless of the prevailing market rate. The guaranteed minimum return is defined by the technical interest rate, see for example Švec (2011). The guarantees often only apply in extreme cases of poor investment returns (Dickson et al., 2009, p. 431). But their presence is significant. Holding no extra capital under the wishful assumption that the guarantee will never apply is a dangerous oversight.

The embedded option of the interest rate guarantee is costly for the insurer when interest rates drop below the technical interest rate. The cash outflows become stuck at the technical interest rate and constant liability cash flows are projected – a constant rate is credited to the policyholders' funds – regardless of the plummeting rates. Following a rate drop, the lower discounting rate and the flooring effect of the guarantee on the value allocated to the policyholder's fund would sharply increase liabilities.

Inadequate management of risk arising from these guarantees has historically led to severe financial problems, such as those of the British company Equitable Life in the 1990s. Similarly dire consequences could arise from the conflation of Macaulay duration with effective (option-adjusted) duration. Because the cash outflows depend on interest rates, it is incorrect to use duration the same way as with fixed-income instruments (bonds); Macaulay and option-non-adjusted modified duration formulas require market-independent cash flows.

3.5 Statutory Reserves vs. Best Estimate of Liabilities

Calculating reserves corresponding to ${}_tV$ by the *statutory approach* locks in the assumptions that were used to calculate the premium at or before policy inception, also known as the *first order assumptions* (Janeček, 2006). This means that for the policy's lifetime, we hold at our inception-day rates the technical interest rate, mortality, and expenses. Lapses are typically assumed as none. Any future development deviating from first-order assumptions may be a source of loss or profit. If the deviation is profitable, such as the investment return on reserve being above the technical interest rate, part is allocated to participating policies. The

assumptions attempting to underpin the expected future development are the *second-order assumptions*. For example:

- non-zero lapse assumptions can be estimated from a sound interpretation of the portfolio's historical records,
- mortality tables are adjusted to fit the insurer's experience and anticipated changes, such as a temporarily reduced life expectancy due to COVID-19 in and after 2020,
- in place of the constant technical interest rate, the insurer may use a deterministic yield curve or assume a stochastic short-rate model,
- expenses are no longer those assumed during premium calculation, but reflect the insurer's latest knowledge.

The Solvency II regulation dictates analysis of past policyholder behavior and best estimates accounting for demographic developments as well as portfolio characteristics (European Commission, 2015, pp. 34–37). Valuation at Solvency II-consistent assumptions produces the *Best Estimate of Liabilities* (BEL). Burkhart (2018) equates BEL to the expected present value of all future cash inflows less the outflows required to settle the insurance obligations. These should be discounted using the risk-free curve published by EIOPA (2025) monthly. A crucial component of BEL is the time value of financial options and guarantees (TVFOG) as discussed in 3.4. This quantity is added to the *certainty equivalent*. We obtain a certainty equivalent by transforming uncertain cash flows into a single deterministic figure that is equivalent in utility to the given set of uncertain cash flows themselves (Ingersoll, Jr., 1986, p. 15). Specifically, the certainty equivalent corresponds to the most expected scenario of market and policyholder behavior development. On top of this base scenario, the TVFOG term accounts for exposure to random fluctuations in market conditions. The valuation of TVFOG is often done by simulation of hundreds of future interest rate development scenarios. Denoting $BEL^{[CE]}$ as the certainty equivalent, we can write

$$BEL = BEL^{[CE]} + TVFOG. \tag{3.13}$$

See further decomposition of BEL in the work of Burkhart (2018). For a demonstration of liability valuation using simulation of interest rate scenarios, see Janeček (2006).

The Solvency II framework denotes the term *technical provision* TP, which is the sum of BEL and the *risk margin* RM. RM increases the technical provisions to the level of the transfer amount, meaning it brings TP up to the market-consistent value a third party would require to assume the TP. Solvency II prescribes that risk margin be calculated using the cost-of-capital method, which depends on the insurer's Solvency Capital Requirement. For further reading, refer to Institute and Faculty of Actuaries (2016).

4 Elements of Asset and Liability Management

One might think that incorporating a risk department reduces the profitability of a business by mitigating rewards from taking on healthy risks. If the said risk is interest-rate risk, one might assume that managers can predict the future movement of interest rates anyway, and there is no need for an extra department. This is, of course, a misconception. If the manager had such thorough market foresight, it is unreasonable to assume they would display the indefinite altruism to continue sharing it with the business rather than generate private personal wealth for their own needs (Fabozzi & Konishi, 1996, p. 9). A risk management department in a life insurance office ensures that it acts as an efficient financial intermediary, which generates profit. The essence of financial intermediation is accepting liabilities to fund assets.

4.1 A Bank's Collapse

A large portion of literature on ALM pertains to banks. If a financial intermediary wishes to hold an asset to maturity, it does not have to recognize price declines due to interest rate hikes in their accounting balance sheet. Instead, the amortized book value in the balance sheet develops according to a rate determined at acquisition, approaching its principal with approaching maturity, but does not respond to interim market rate shifts. We call this a measurement at *amortized cost* rather than at market value. The Silicon Valley Bank (SVB) valued their bonds by this method. Over 2020 and 2021, the SVB decided to invest around \$91 billion in Treasury bonds and mortgage-backed securities (Jha et al., 2025). SVB's investments in stocks, bonds and other securities then amounted to over half of the bank's assets rather than the quarter typical among its peers. Crucially, to achieve the highest yields, the assets' maturities were often in double digits, implying also high duration, and hence high sensitivity to rate shifts. Of all the investments, SVB designated 75% as held-to-maturity (HTM), justifying the use of valuation at amortized cost, which does not adjust according to rate changes. Meanwhile, the US Federal Funds Rate range jumped from 0.00–0.25% in March 2022 to 4.50–4.75% by February 2023 (Federal Reserve Bank of New York, n.d.). The amortized cost valuation may have made the impact of the rate hike easy to overlook. But as depositors withdrew funds, the bank had to liquidate all its available-for-sale assets and had no choice but to also eventually sell parts of its HTM portfolio. Of course, as these assets

were long duration, they had accumulated deep hidden unrealized losses after the rate hike. The bank's forced liquidation of HTM assets played a major role in the bank's collapse in March 2023, marking the second largest in US history. Although fueled by panic in an ever more fast-paced world, we see the fall of SVB partly as an asset-liability management failure in practice.

The predominant difference between the asset-liability composition of insurance companies and banks is that a bank holds longer-dated assets than liabilities. A bank accepts cash deposits which may be withdrawn at short notice while issuing loans and mortgages – it borrows short to lend long. The profile is most often reversed for an insurance company, as their liabilities contain long-term policies or annuity payouts, while long-dated bonds are often unavailable. Ho and Schneeweis (1999, p. 24) specify that a good ALM strategy considers the interest rate risk, liquidity risk, and default risk. Default risks for a bank stem primarily from the loans it has issued. Insurers face default risk arising not only from their investment portfolio, but, somewhat metaphorically, also from the danger that death or catastrophe – an insured event – abruptly stops the stream of premiums collected, which motivates cash flow modelling and scenario testing.

4.2 Manipulation of Insurer's Assets and Liabilities

The greatest part of a life insurer's liabilities are the technical provisions, typically divided into life- and non-life-insurance-related technical provisions. These are determined by the cash flows arising from the contracts the insurer has entered. The limited options of the insurer to change the liabilities' side of the balance sheet lie in:

1. deciding what products are offered at the outset,
2. undertaking actions to alter policyholder behavior. For example, introducing and marketing a new product may alter the lapse rates used to calculate the total value of liabilities and thus technical provisions.

The first item on the list above cannot be exercised once the product is sold. But a specific aspect of life insurance contracts is that once settled, they may have long maturities, such as 50 years. This leaves the second approach, which is non-standard. Manipulating the assets' side of the balance sheet is more feasible.

The largest item on the assets' side tends to be government bonds. This is the case for ČSOB Pojišťovna – see its report on solvency and financial condition from 2023 (ČSOB Pojišťovna, 2024, p. 13), which identifies the Czech government bonds as the dominant part of its asset portfolio. Similar conclusions regarding debt instrument preference can be made after scanning the 2023 annual reports of other major Czech insurers, such as Generali Česká

pojišťovna and UNIQA pojišťovna, where fixed-income securities dominate as well. The remaining portion is, if not in cash or deposits, diversified across less conservative assets such as equities, real estate, etc. One of the primary goals of asset liability management can be seen as manipulation of the assets' side such that its characteristics most closely resemble or match the liabilities' side's. A crucial part of a company's net worth is the value of its assets minus the value of its liabilities. We see then that matching these characteristics on both sides could preserve the net worth by protecting it from market-driven shifts. Shareholders find this desirable, as they wish to see the company's net worth grow with reasonable volatility.

4.2.1 Shifts in Liabilities

The value of a company's liabilities can change because of a shift in any of the assumptions stated in 3.2:

1. the insured population submits more claims than predicted, e.g., due to changes in mortality,
2. the insured cancel their policies at levels other than predicted,
3. the insurer incurs unexpected costs, for reasons ranging from miscalculation to unforeseen inflation and others.

Furthermore, the value of liabilities can unexpectedly shift due to:

4. catastrophes – such as natural disasters – which lead to unexpectedly high claim numbers,
5. operational risk – human errors, technological malfunctions, hacker attacks, and fraud.

The above five constitute the *non-financial risks*. An observant reader notices that the interest rate assumption was so far omitted, as interest rate shifts belong to financial risks instead, and affect not only the liabilities, but also the assets.

4.2.2 Shifts in Assets

There are factors beyond interest rates influencing the value of the assets chosen for investment. For example, to mitigate credit risk, ALM may seek to control the bond composition. It may be required that no more than a specified percentage of the bond portfolio be invested in bonds rated BBB and lower. Similarly, we may limit the volume invested in bonds of certain types and sectors – municipal, government, issued by financial or energy companies – not to exceed a certain percentage of the total, to ensure diversification.

Potential losses in equity and real estate can be limited by ensuring that the 1-year Value-at-Risk of the portfolio expressed as a percentage of the total investment asset value does not exceed a specified threshold. If multiple currencies are present in the balance sheet, tolerance of foreign exchange risk should be decided – oftentimes as zero tolerance of risk – and can be hedged by swaps and forward contracts (Janeček, 2024).

4.2.3 Shifts due to the Interest Rate

In practice, we say that the interest rates shift if yields change on the instruments selected in risk-free interest rate calculation – most often government bonds and interest rate swaps – and such events are out of reach of managerial influence. The risk-free interest rate is then used for discounting both asset cash flows and liability cash flows. Also, profit sharing, if present, links the liability cash flows to asset performance, which ties back to interest rates.

4.3 The C-risks

The attempt at protection against interest rate shifts can be broadly understood as addressing the C-3 risk, one of the four actuarial “C-risks” (C standing for Contingency) as coined by Charles L. Trowbridge (Panjer, 1998, p. 93):

1. C-1: asset quality risk—the risk of asset defaults and of decreases in value of equity investments.
2. C-2: insurance pricing risk—risks due to potential increases in claims resulting from, for instance, inaccurate assumptions of mortality, lapses, or expenses and other pricing deficiencies.
3. C-3: interest rate risk—the risk of losses resulting from a change in the level of interest rate or the shape of the yield curve.
4. C-4: business risk—an umbrella category for accounting, managerial, social, regulatory risks and perhaps all which cannot be fit in the previous C-risks. Operational risk falls in this category. Maier (2004, p. 28) mentions market risk from business expansion, changes in taxation and regulation, insurance fraud, mismanagement and lawsuit risk.

The implication of the C-3 risk is not that assets and liabilities are to be matched perfectly, but rather that the dangers and consequences of asset-liability mismatch are considered (McCarthy et al., 1982, p. 357).

4.4 Static techniques

By the 1980s, American financial institutions controlled both the interest rate risk and liquidity risk by the *maturity bucket approach* (Brewer, 1985). Grouping assets and

liabilities into different maturity buckets, we can match the cash flow totals of assets to cash flow totals of liabilities for each bucket. Drawbacks include that the time buckets themselves are set arbitrarily. For example, one bucket comprises all cash flows occurring within 0-12 months from now. We also assume that the interest rate changes uniformly within each bucket, and the method neglects embedded options.

The above is an example of *static techniques* as outlined in the work of van der Meer and Smink (1993). Static techniques differ from *dynamic strategies* in that the former does not embed ongoing decision-making rules for behavior. An output of a particular technique may reveal mismatches and serve as basis for decision-makers to restructure the portfolio. Of static techniques, Meer and Smink mention:

- Cashflow payment calendars,
- Gap analysis,
- Segmentation,
- Cash flow matching.

4.4.1 Cash Flow Matching

The example of maturity bucket matching we saw earlier can be formalized into cash flow matching, which uses linear programming. In their book, Fabozzi & Fabozzi (2021, Chapter 27) show cash flow matching, also calling it a *dedicated portfolio*, on the following algorithm:

1. Identify the final unmatched liability cash flow.
2. Select a bond whose maturity aligns with the final liability cash flow. Invest such that the sum of the bond principal and the final coupon equals the liability cash flow.
Once matched to an asset, consider the final liability covered – matched.
3. Reduce all the previous – earlier – liability cash flows by the coupon payments of the bond just obtained.

The above process is repeated until all liabilities are matched. Elton et al. (2014, p. 585) define the matching problem as follows.

Given the elements:

1. $L(t)$ as the liabilities in time t ,
2. $C(t, i)$ as the cash flows in period t from a bond of type i ,
3. $P(i)$ as the price of bond i ,
4. $N(i)$ as the number of bonds of type i purchased,

and the constraints:

1. The bond asset cash flows suffice to meet liabilities:

$$\sum_i N(i)C(t, i) \geq L(t) \quad \forall t,$$

2. The investor cannot issue bonds:

$$N(i) \geq 0 \quad \forall i,$$

we

$$\text{minimize } \sum_i N(i)P(i),$$

(4.1)

which completes the formulation of the matching problem.

Recall that the asset cash flow consists of income generated by the investments, as well as any principal repayments when an investment matures. Liability cash flow is the sum of policy outgo (such as claims, policy surrenders, expenses, ...) less income from premiums. Consider the net cash flow during the year t , $CF(t)$, as the difference between asset cash flows and liability cash flows in year t . We can recognize two components to the interest rate risk itself:

1. Reinvestment risk—occurs when $CF(t)$ is positive; there is a risk that the excess income be reinvested at an interest rate lower than forecast.
2. Disinvestment risk (price risk)—occurs when $CF(t)$ is negative; to cover the cash shortage, assets may have to be liquidated. There is a danger that this must be done at undesirably low values if the interest rate has risen.

Notice that the approach in (4.1) eliminates disinvestment risk. But it allows $CF(t)$ to be positive, hence ignoring reinvestment risk. This is because exact cash flow matching is impractical. To address both risks while avoiding exact matching, we modify the first constraint in (4.1) to allow flexibility bands bounded by $\Delta_1(t)$ and $\Delta_2(t)$:

$$L(t) + \Delta_1(t) \geq \sum_i N(i)C(t, i) \geq L(t) - \Delta_2(t) \quad \forall t.$$

If the cash flows are collapsed into buckets, i.e.,

$$C(b, i) = \sum_{t \in b} C(t, i),$$

we can express the constraint as

$$L(b) + \Delta_1(b) \geq \sum_i N(i)C(b, i) \geq L(b) - \Delta_2(b) \quad \forall b,$$

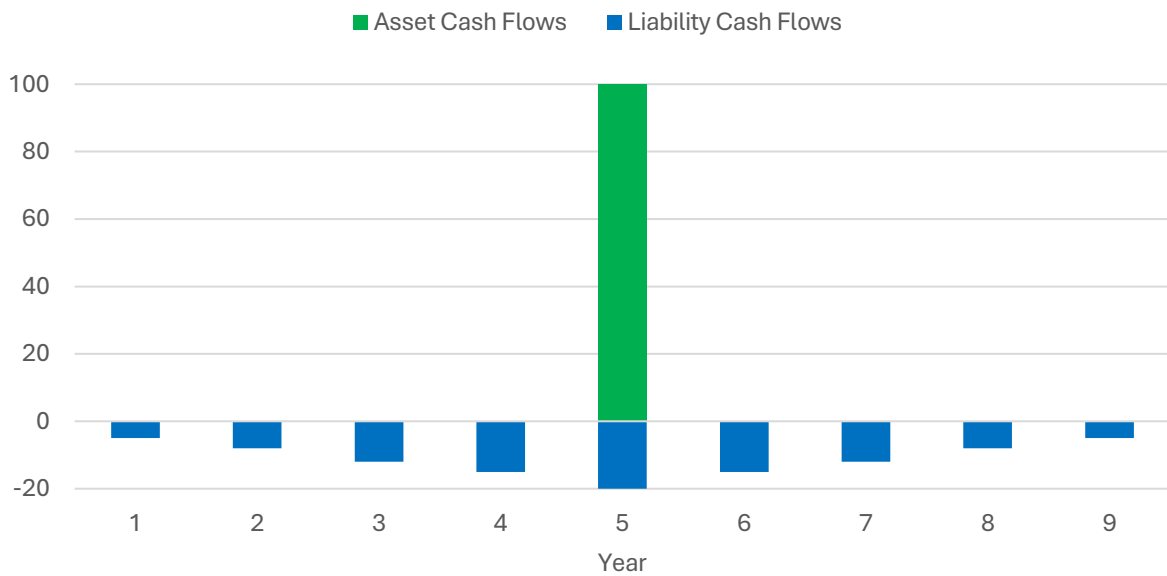
(4.2)

where $L(b)$ is obtained by summing the liability cash flows over all times t belonging to the bucket b . The bucketing approach is illustrated in Figure 4.2.

4.4.2 Practical Cash Flow Matching Scenario

Figure 4.1

Inappropriately matched liability cash flow series



Note. Adapted from Janeček (2024).

We show the usefulness of cash flow matching in this subsection, crediting M. Janeček (personal communication, February 2025). Observe Figure 4.1, which captures the asset and liability cash flows from year 1 to year 9. Such an asset cash flow might arise from a 5-year zero-coupon government bond. The liabilities cash flows were chosen for illustrative purposes so that their Macaulay duration is approximately 5, which matches the asset's profile. In fact, the duration of assets matches that of liabilities exactly if we assume an

interest rate of $r = 0\%$. The arising risk is instead connected to funding the liability cash flows in years 1 to 4 and 6 to 9.

Because the bond is government-issued, there are few to no liquidity concerns – in years 1 to 4, we may fund the liability by selling the bond incrementally. What is more important is the arising disinvestment risk – price risk. Suppose the liability cash flows do not respond to interest rate shifts or at least, as is very often the case, respond to them differently than the assets even despite matched durations – be it due to convexity mismatch or assumption violations. There is a danger that the bond has fallen in market value at any one or each of the times we choose to liquidate it by parts. In unfavorable conditions, come year 5, we will have liquidated a greater portion of the bond than expected to cover the liabilities, and less will be available to cover the future liability cash flows.

On the other hand, at the point we receive the bond principal in year 5, there is a need to reinvest the cash to generate interest. Reinvestment risk enters the picture. Say the interest rate had decreased, then prices would have increased – the available securities might suddenly be unexpectedly costly. These instruments then would not generate the necessary value for the insurer to finance the remaining future liability outflows.

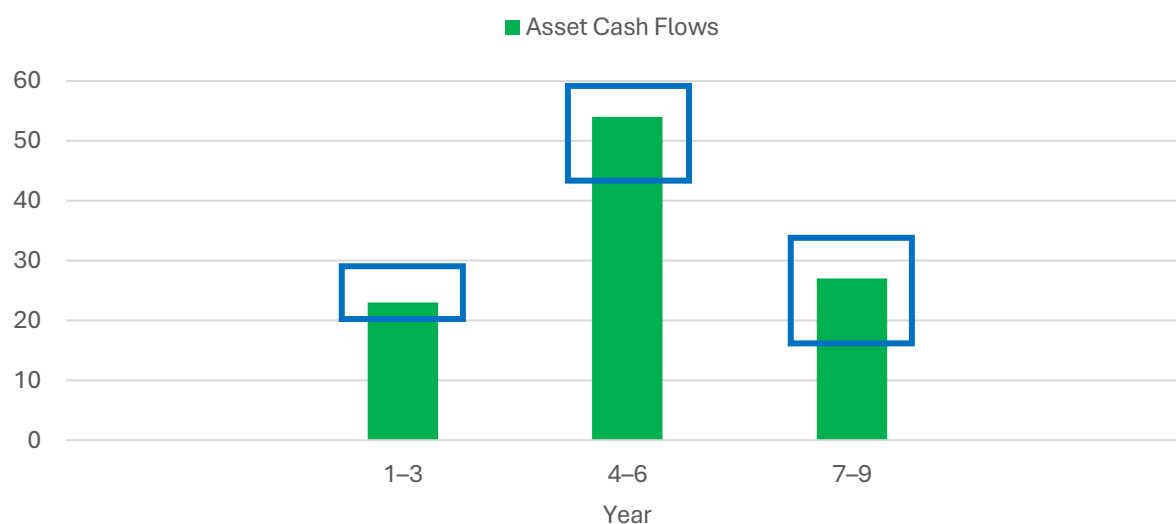
Risk exposure emerges at each point where the asset cash flow is unequal to the liability cash flow. An ideal cash flow configuration would then assign to a liability cash flow an equally sized asset cash flow at each time t . Choudhry (2007) calls this *matched book*. As we primarily consider government bonds, we would need a bond principal repayment at each t . But risk-free investments might be unavailable for maturities over 20 years, while a life insurance contract may easily be active for the next 50 years. So, not all bond maturities will be available in practice, and even if they were, we will see that attempting for an exact match at each t is good for only as long as the value of liabilities remains unchanged. Any emerging gap is dynamic – its size is always a mere snapshot in time and changes daily with new commitments on both the asset side and liability side. And as soon as the liability values change due to assumption shifts, portfolio rebalancing with an aim for exact cash flow matching would repeatedly generate transaction costs and costs related to realized capital gains, such as tax filing and compliance. Accordingly, it is reasonable to allow for flexibility rather than pursue exact matching.

For example, we may collapse liability cash flows in Figure 4.1 into three buckets: years 1 to 3, years 4 to 6 and years 7 to 9. In each bucket, we estimate the liability value and draw flexibility margins above and below the estimate – the further in the future the cash flow bucket, the greater the allowed flexibility to account for uncertainty. Then, we attempt to fit

the total asset-related cash flows in that period inside the interval defined by the margins, as illustrated in Figure 4.2.

Figure 4.2

Flexible maturity bucket matching



Note. Adapted from Janeček (2024).

Consider that an insurer's asset portfolio is diversified and includes equities and real estate rather than bonds only. We can think of stocks and real estate as strategic positions with expected long-term returns that would cover long-term liability cash flows. On the other hand, cash is suitable for short-term liability funding and together with treasury bills maintain a buffer for the event of a funding crisis.

See Kocherlakota et al. (1988), who make adjustments such as allowing cash carry-forward in case of excess and cash borrowing in case of deficits to meet liabilities. Cash flow matching in the sense of (4.1) has no solution if the liabilities have a longer time horizon than the maturities of available bonds. Iyengar and Ma (2009) use stochastic evolution of interest rates and manage the risk in non-exact matching.

4.5 Immunization

If there were a father of asset-liability management in insurance, it would be the British actuary Frank Redington (1952), who first introduced the theory of immunization. This was

later recognized as one of the value-driven dynamic strategies as referred to by van der Meer and Smink (1993). Let us recall the Taylor approximation in (2.5):

$$P(r + \Delta r) = P(r) + \frac{dP(r)}{dr} \Delta r + \frac{1}{2!} \frac{d^2 P(r)}{dr^2} (\Delta r)^2 + \dots$$

Consider an analogy where in the place of $P(r)$ we have the net worth of a company defined here as the difference between the present value of asset-related cash flows (asset-proceeds) V_A and the present value of liability-related cash flows V_L (liability-outgo), i.e. the net worth is $(V_A - V_L)$. Further, let us granularize compounding infinitesimally to transition to the force of interest δ defined in (1.5). If we also suppose a change in the force of interest from δ to $\delta + \varepsilon$, we can easily progress from the formula (2.5) seen above to:

$$(V'_A - V'_L) = (V_A - V_L) + \frac{d(V_A - V_L)}{d\delta} \varepsilon + \frac{1}{2!} \frac{d^2(V_A - V_L)}{d\delta^2} \varepsilon^2 + \dots , \quad (4.3)$$

where $(V'_A - V'_L)$ is the net worth after the shift in force of interest – this is how Redington (1952, p. 4) begins his reasoning. We let $V_A = V_L$ so that the first term of (4.3) is zero. Redington characterizes “matching” as aligning asset cash flows to liability cash flows so that their vulnerability to exogenous changes matches. He coins the word immunization to specifically denote such a form of matching that the investor’s (company’s) net worth is protected against interest rate shifts. We begin attempting this by matching the discounted mean term, i.e. duration, of assets to that of liabilities, yielding Redington’s first condition for a satisfactory immunization policy:

$$\frac{d(V_A - V_L)}{d\delta} = 0 , \quad (4.4)$$

which sets the second term of (4.3) equal to zero. Notice that as long as the second derivative is positive, the term $\frac{1}{2!} \frac{d^2(V_A - V_L)}{d\delta^2} \varepsilon^2$ in (4.3) will also be positive, increasing the net worth regardless of how δ moves. Recall Section 2.3 where we discussed how positive convexity indeed always benefits the holder of the security. Hence Redington’s second condition in immunization:

$$\frac{d^2(V_A - V_L)}{d\delta^2} > 0 . \quad (4.5)$$

4.5.1 The Limitations of Immunization

A problem with standard immunization is the requirement of flat yield curves; Reitano (1991) overcomes some deficiencies of Redington's immunization by expanding to key rate shifts across multiple maturities. Furthermore, immunizing the portfolio while keeping the convexity of assets higher than that of liabilities creates an arbitrage opportunity. Under the parallel shift assumption, if rates fall, assets should appreciate more steeply than liabilities, and if rates rise, the losses on the liabilities' side are higher than on assets' side. Any parallel yield curve shift will thus produce an inevitable net gain in the portfolio.

Disappointingly, when building its investment strategy, an insurance company may discover that bonds and fixed-income instruments do not provide as much convexity as its liabilities do (Ostaszewski, 2002, Section 3). Trieu (2013) notes that in a rate drop, guarantees increase liability duration, while in a rate hike, early surrenders reduce liability duration. In the case of Trieu's and Ostaszewski's observations, the net worth has negative convexity. This leads to the opposite of the portfolio gain discussed above – any parallel shift of the yield curve will cause a net loss to the portfolio. Ostaszewski illustrates the weakness of immunization on the sharp interest rate downturn in the United States over 1991 and 1992. In that scenario, the rate drop makes the immunized net worth decline due to negative convexity of the net worth. The insurer, committed to duration matching, immunizes the portfolio in the new low-rate environment again, only for the net worth to decline again once the rates move back up substantially by 1995. The author argues for a greater acceptance of duration mismatch and treats the balance sheet and net worth as derivative securities.

4.6 ALM Implementation

The insurer encodes their asset liability management practices in a document called the *investment strategy*. Its contents are decided broadly by the Board of Directors, who are elected by the shareholders. The objectives of the strategy are typically to maximize investment performance while remaining within approved risk parameters. The strategy is implemented by the chief financial officer (CFO) and the chief risk officer (CRO). Investment performance not only determines profit, enabling higher dividend payments to shareholders, but also the profit-sharing payouts on participating policies. The *risk appetite statement* states risk targets. Of note is maintaining a particular insurer financial strength rating assigned by the major rating agencies – Standard & Poor's, Moody's, Fitch, and A.M. Best. For instance, Generali states the A rating from A.M. Best in their 2023 report on solvency and financial condition (ČSOB Pojišťovna, 2024). Refer to Rejda et al. (2022, p. 297) for the exhaustive list of ranks. Other targets may include a high enough Solvency II ratio – the ratio of own funds to the Solvency Capital Requirement. There may also be a limit on the worst

case modelled one-year economic loss – be it a historical scenario or a one-year 99% VaR – to at most one year’s net income, etc. The investment strategy segments the asset portfolio into portions (buckets) and specifies the correspondence of these portions to different types of liabilities. Further specified are the necessary actions undertaken if moving assets between portfolio portions – for example, if assets are removed from the portfolio bucket backing profit-share policies, the insurer duly allocates any realized gains to the corresponding profit-share policies. After defining key terms, the document imposes limits for each portfolio portion, summarized as we outline them below in Subsections 4.6.1–4.6.4. Janeček (2024) recommends that these metrics be monitored in ALM-related reports.

4.6.1 Interest rate risk

$$10bpsBPV_{Assets} - 10bpsBPV_{Liabilities} < \text{specified \% of invested assets}; \quad (4.6)$$

the 10-point basis point value gap between assets and liabilities – equivalent to the 10-point BPV of own funds – does not exceed a specified percentage of the invested assets.

$$DG = D_{E,Assets} - \frac{L}{A} D_{E,Liabilities} < \text{specified threshold}, \quad (4.7)$$

where L is the value of liabilities and A is the value of assets; the (effective) *duration gap* does not exceed a specified threshold.

$$KDG = KRD(i)_{Assets} - \frac{L}{A} KRD(i)_{Liabilities} < \text{specified threshold } \forall i; \quad (4.8)$$

the key rate duration gap does not exceed a specified threshold for each maturity i .

$$\Delta Own Funds < \text{specified amount for each stress scenario}; \quad (4.9)$$

where the stress scenarios tested may be solvency II-related, the New York 7, or historical stresses.

4.6.2 Reinvestment and Disinvestment Risk

$$L(b) + \delta_b L(b) > CF_{Assets}(b) > L(b) - \delta_b L(b) \quad \forall b; \quad (4.10)$$

cumulative asset cash flow in each maturity bucket b is within $\delta_b \cdot 100\%$ of the liability cash flow for maturity bucket b .

4.6.3 Asset Limits

For bonds, we may set limits for ratings:

$$\% \text{ of bonds rated } B \text{ and lower} < \text{specified \% of invested assets}, \quad (4.11)$$

and similarly for sectors and issuing countries. For stocks,

$$VaR_{1Y,99\%,stocks} < \text{specified \% of invested assets}, \quad (4.12)$$

and we may impose stock sector limits as for bonds. Separately for real estate,

$$VaR_{1Y,99\%,RE} < \text{specified \% of invested assets}. \quad (4.13)$$

To ensure liquidity,

$$\% \text{ of highly liquid assets} > \text{specified \% of invested assets}. \quad (4.14)$$

4.6.4 Other Limits

There is normally no foreign exchange risk allowed. Derivatives are rarely used by insurers; if used, they may only be allowed to hedge foreign exchange risk (Janeček, 2024). Limits on intra-group and out-of-the-group assets ensure there is no excessive reliance on a single fund source during stress events.

4.7 The ALCO

Once the technical requirements for ALM are satisfied and key metrics pertaining to risks are calculated, the decisions on how these risks are managed may be reserved for the *Asset and Liability Committee*, abbreviated as the *ALCO*. The extent of authority of the ALCO differs among insurers – some ALCOs merely review and discuss mismatches and compliance with regulation to forward insights to higher management, others are formal decision-making

bodies (Deloitte Development LLC, 2020). Federal Reserve System (n.d.) encourages quarterly meetings, and indeed the 2020 Deloitte report saw 40% of the 18 surveyed insurers report ALM metrics quarterly, while 30% reported monthly. In literature, this committee is most often discussed within bank ALM, see Choudhry (2007, Chapter 8) or Choudhry (2020), who advocates monthly convening.

The ALCO recruits broad viewpoints from Investments, Capital Management, Risk Management, Actuaries and is chaired by the CFO. The asset manager makes the day-to-day trades, while actuaries maintain and ensure accurate cash flow models. An ALCO meeting may begin with the asset manager commenting on the economic outlook and development, the political situation, and in particular yield curve expectations. The CFO reviews the asset manager's realized investment return in relation to the benchmark and checks the asset portfolio against predefined budgeting. The CRO reports on each risk metric and whether they stay within limits. Buy/sell recommendations are formulated to avert or prevent risk limit breaches, e.g., to ensure enough liquid cash (Janeček, 2024).

4.8 More Balance Sheets

Insurers conduct ALM on multiple balance sheets simultaneously. In the discussion so far, we have considered the *economic balance sheet*, tracking cash flows discounted by the risk-free yield curve according to actuaries' internal models. In the economic balance sheet, we follow the best estimates and market values by definition, hence calling the approach *market-consistent*. The values of assets are quoted on the market directly, while liabilities are valued by discounting probability-weighted future cash flows.

Liability valuation in the *regulatory balance sheet* (Solvency II balance sheet) excludes any cash flows occurring after the contract boundary date. For example, if the life insurance policy contains an accident add-on that the insurer themselves may terminate, this liability cannot be modelled past the point of the earliest possible termination. Solvency II prescribes the yield curve at which cash flows are discounted – to be found on the EIOPA (2025) website – and the liabilities include the risk margin as seen in Section 3.5. We may monitor and stabilize the solvency ratio – the ratio of own funds to the Solvency Capital Requirement.

The methodology used by SVB, as outlined in Section 4.1, valued assets at their amortized cost, deviating from market consistency. This is a feature of a *reporting balance sheet*, or alternatively the accounting/statutory/P&L balance sheet. Here, we can track the sensitivity of profit/loss, which influences dividends paid to shareholders. The risk adjustment differs from Solvency II's risk margin, and unrealized profits or losses may be deflected to contractual service margin (CSM) and other comprehensive income (OCI) under IFRS17. Priorities are assigned as stabilizing earnings and regulatory metrics may undermine the original goals of duration and cash flow matching.

5 Multiscenario Analysis

In this chapter, we analyze an insurer's portfolio's response to multiple interest rate shifts. The liabilities' portfolio is composed of four commonly traded types of insurance products:

1. life insurance with profit sharing (participating contracts),
2. life insurance risk policy products,
3. life insurance unit-linked products,
4. non-life insurance products,

further commentary on which we provide below. For the purposes of the work, we modified a liability cash flow model prepared by RNDr. Martin Janeček, Ph.D., with his kind approval. In the asset portfolio, we model fixed-coupon bonds. We use the Czech crowns as currency units. The current valuation year is set to 2025 as we model yearly cash flows. We use the 2023 life tables for mortality data (Czech Statistical Office, 2024) and consider the mortality rate of the insured population to be half that of the general population and constant annual lapse rates of 20%. Unless noted otherwise, the figures and tables in this chapter are own work generated in Microsoft Excel and RStudio v4.2.2 (RStudio Team, 2020) using the ggplot2 package v3.5.0 (Wickham, 2016). Let us isolate the insurance products to see each product's scenario-based duration profile and how it compares to bonds.

5.1 Life Insurance with Profit Sharing

These contracts contain both a pre-set interest rate guarantee and profit sharing. Liability cash flows are pooled, invested, and a share of the investment return is then allocated to policyholders' funds. The benefit paid in case of death while the policy is active amounts to a fixed sum assured, which we set to $SA = 50\,000$ Czech crowns, plus the accumulated fund value. In the case of survival to maturity, the policyholder receives the accumulated fund value. In case of surrender, the policyholder receives the accumulated fund value reduced by a surrender charge.

In our model cases, we guarantee a rate of 3%, while we assume an investment return and a discounting rate of 5%, implying a flat term structure. Assume a male policyholder of age $x = 40$ who had just been issued a participating policy with 10 years remaining until its expiration. We set the premium so that the present value of future expected liability cash

flows is zero, giving $P = 4\,092.90$ Czech crowns yearly. For the specified single policy, the fund accumulates as seen in Table 5.1.

Table 5.1

Fund value accumulation for one participating policy

t	$Fund_t$
1	4 193.34
2	8 530.12
3	13 014.69
4	17 651.38
5	22 444.50
6	27 398.27
7	32 516.76
8	37 803.94
9	43 263.73
10	48 899.96

Note. $Fund_t$ denotes the end-of-year fund value.

Let us assume a portfolio of 500 000 such participating policies on the liabilities' side. On the assets' side, we model 85 000 bonds with an annual coupon, $F = 10\,000$, $c = 0.05$, $T = 10$ years, such that the greatest liability cash flow is complemented with a bond principal repayment, as seen in Figure 5.1. We plot the changes of assets and changes of liabilities in response to different interest rate shifts in Figure 5.2.

Figure 5.1

Cash flow alignment, single maturity profit sharing

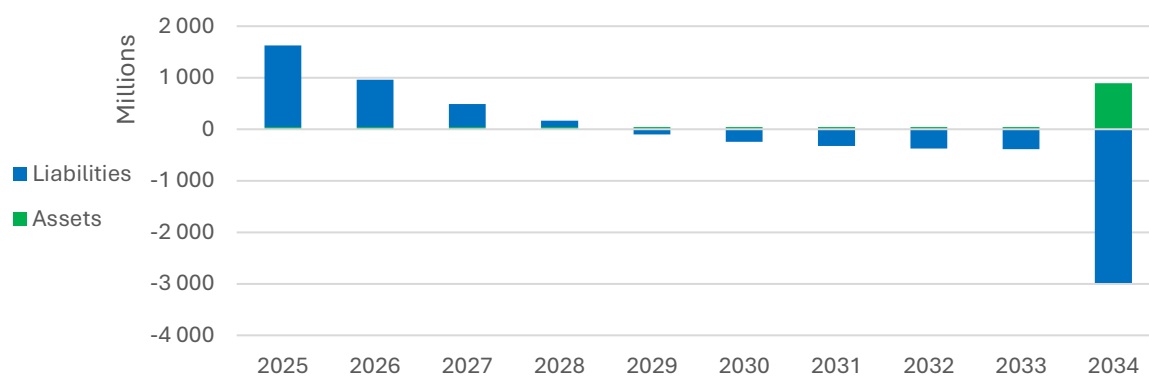
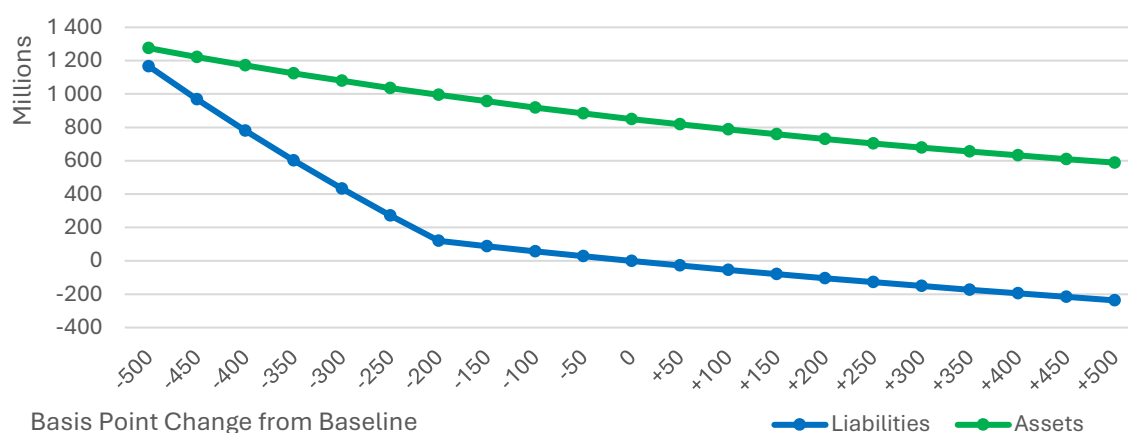


Figure 5.2

Asset & liability responses to ± 500 bp parallel shifts, single maturity profit sharing

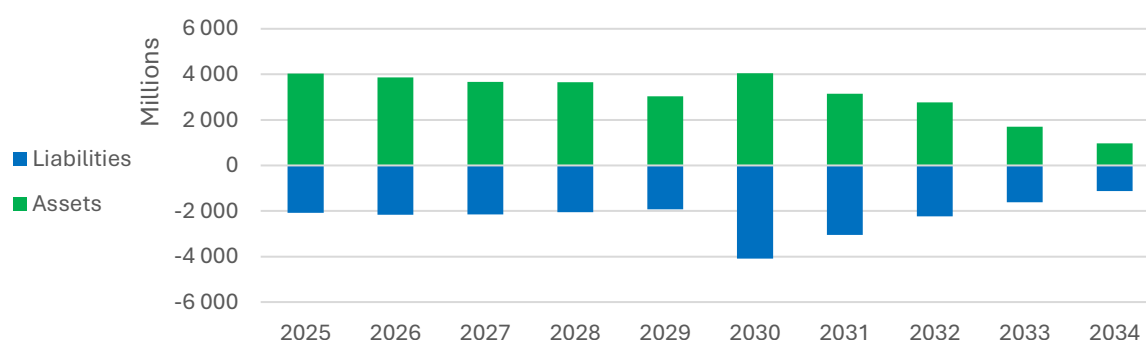


Note. Premium was set such that the policy value is modelled as zero at baseline interest rate at policy inception.

Before we comment further, suppose a modification where we have five types of profit-share policies, such that there are 6, 7, 8, 9 and 10 years left until policy expiration, 100 000 of each type. Each policy type was issued with a period of 15 years to 55-year-old women in years 2016 to 2020, while the reference year remains 2025. We modify the portfolio of bonds such that the greatest bond principal repayments are aligned with greatest liability cash outflows, as illustrated in Figure 5.3.

Figure 5.3

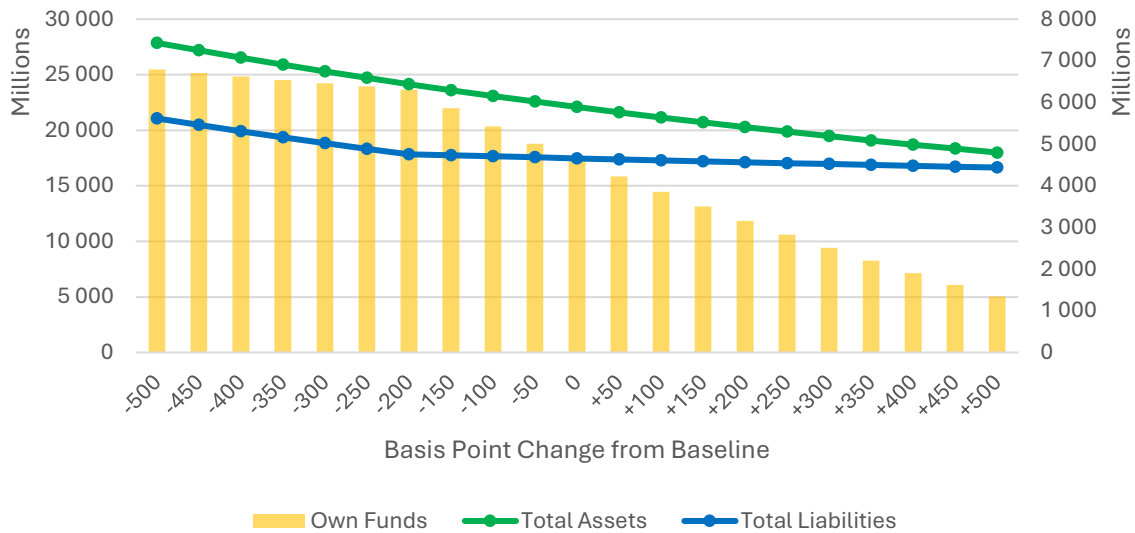
Cash flow alignment, profit sharing policies of several maturities



We capture the sensitivity of market values of assets and liabilities under different interest rate shocks, this time while tracking the value of own funds – the company surplus/equity – in one place using a secondary axis on the right in Figure 5.4.

Figure 5.4

Asset, liability, and own funds responses to ± 500 bp parallel shifts, profit sharing policies of several maturities



Note. Own funds displayed on the secondary axis.

Notice the commonalities in Figure 5.2 and Figure 5.4. The market value of assets, in green, declines in both cases predictably with a rising interest rate. Rather than declining linearly, the trend curves slightly, which results from convexity described in Section 2.3.

The market value of liabilities, in blue, increases with a decline in interest rates. As the interest rate declines from baseline, the projected liability cash flow rises slightly due to lesser impact of the discount factor $v = \frac{1}{1+i}$. However, note the kink in the liabilities' curve, which occurs at the parallel yield shift of -200 basis points. The change in slope is a consequence of the option embedded in the liability, namely the interest rate guarantee. The kink at the point -200 basis points occurs because we had set the interest rate guarantee at 3%, while the base scenario of no rate shift, i.e., the value at the 0-point on the horizontal axis in both figures, corresponds to a rate of 5%. From the shift of -200 basis points onward to the left:

- the cash flow projected ceases to depend on interest rates – constant additions to policyholders' funds are projected, determined by the guarantee;
- in Figure 5.2, as liabilities increase in value at a faster pace than the assets do, own funds are diminished.

Own funds may decline dramatically also as the interest rate increases, as seen on the right side of Figure 5.4. This results from the profit share feature. Without profit share, the

liabilities would decrease more sharply as the interest rate rose, due to the increasing effect of the discount factor $v = \frac{1}{1+i}$. But in our model, we allocate to the fund 80% – the profit share ratio – of the excess investment return obtained beyond the guarantee, which offsets the discounting effect, and the values of liabilities do not fall as sharply. Resultingly, the rate at which the assets fall is greater than the rate at which liabilities do, and equity shrinks as interest rate rises. Recall that taking the negative of the rate of change of assets' or liabilities' market value with respect to the interest rate and dividing by their market value approximates duration. Calculating scenario-based effective duration by $D_E \approx$

$-\frac{P(r+\Delta r)-P(r-\Delta r)}{2\Delta r P(r)}$, where we let $\Delta r = 0.005$, gives the values in Table 5.2, plotted in Figure 5.5.

We defined the duration gap as $DG = D_{E,Assets} - \frac{L}{A} D_{E,Liabilities}$ in (4.7), where the factor $\frac{L}{A}$ normalizes durations to assets. We show duration gaps in Table 5.2.

Table 5.2

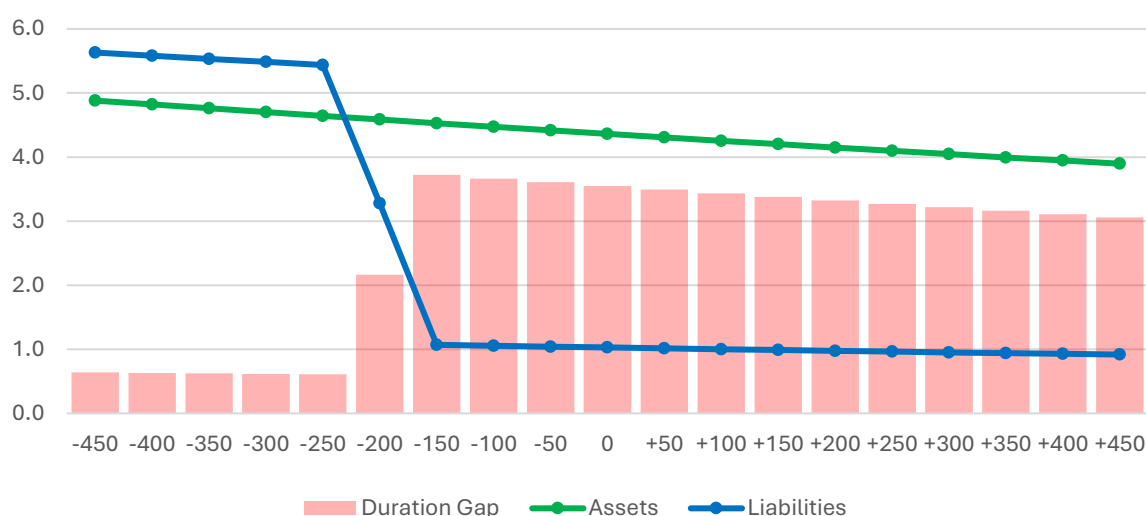
The duration gaps between assets and liabilities for a portfolio of profit-sharing policies of several maturities

Parallel Rate Shift	Assets' Effective Duration	Liabilities' Effective Duration	Duration Gap
-400 bps	4.821	5.582	0.631
-300 bps	4.702	5.485	0.617
-200 bps	4.586	3.280	2.162
-100 bps	4.473	1.056	3.665
0 bps	4.362	1.029	3.549
100 bps	4.254	1.002	3.435
200 bps	4.149	0.977	3.324
300 bps	4.046	0.953	3.216
400 bps	3.946	0.929	3.111

Values in Table 5.2 imply that if the current constant interest rate increased (decreased) by a hundred basis points for every maturity, assets would decrease (increase) in value by 4.362%, while liabilities only by 1.029%. If the rate were 300 bps below today, a parallel move down (up) of the yield curve by a hundred basis points would increase (decrease) the value of assets by 4.702%, and the value of liabilities by 5.485%.

Figure 5.5

Duration of bonds, duration of profit-sharing policies, and the duration gap



Note. Duration of bonds is in green, duration of profit-sharing policies in blue, and the duration gap in pink.

Figure 5.5 and Table 5.2 reveal large duration gaps as rates rise from baseline. It appears from Figure 5.4 that a positive duration gap benefits the insurer's own funds when rates fall. Financial intermediaries may make calculated profits by taking on these risks (D. Ho & Schneeweis, 1999, p. 25). But recall the case shown in Figure 5.2. Beyond -200 bps, the equity quickly deteriorated. ALM seeks to eliminate this vulnerability. In terms of Figure 5.4, we wish to align the slope of the green assets' curve to the slope of the blue liabilities' curve – ideally, the curves would appear parallel to each other.

5.1.1 Aligning Slopes

The insurer is limited in changing the structure of their liabilities; stopping the issuance of participating policies might be infeasible due to market competition. The best hedge against interest rate shifts would be the purchase of tailor-made financial options that have a similar duration profile as the liabilities. However, the specificity of such a financial instrument would make it not only expensive but also illiquid in the Czech market (Janeček, 2024). Furthermore, the duration profile of the liabilities may easily change as model assumptions are reconsidered and adjusted when, for example, the experienced lapse rate increases. The asset portfolio could quickly be rendered outdated, but the insurer would sell the previously obtained financial option most likely at a loss due to low liquidity.

Lowering the profit share ratio when rates rise may preserve equity. If legally permitted, this may be an effective hedge but note it may encourage lapses. We may also dynamically adjust

the bond portfolio to match durations on the assets' and liabilities' sides. The negatives are those that we have already noted for cash flow matching in Section 4.4.2 – continuous rebalancing generates transaction costs and tax costs related to realizing capital gains. The compromise is monitoring the duration gap and taking action once it exceeds a pre-determined limit.

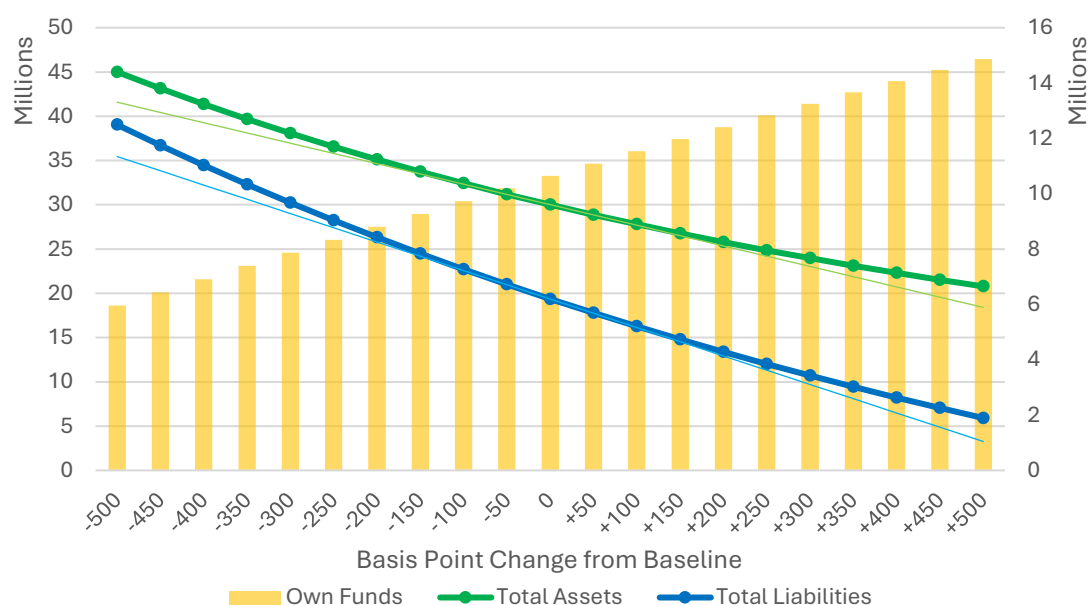
5.2 Life Insurance Risk Policy Products

Term risk products cover only the risk of death. In contrast with profit-sharing contracts, the accumulating fund value is negligible, and none is paid upon surrender or maturity. Assume a portfolio of 500 000 such policies. Each was issued to a 50-year-old male for a policy period of 15 years and has now 10 years remaining until expiry, at the sum assured of CZK 20 000 and a yearly premium of CZK 165.

We complement the liabilities with an asset portfolio of 3 000 bonds of face value 10 000 each and maturity in 10 years. Observe how assets and liabilities adjust in market value to interest rate changes in absence of the option-like characteristics arising from the interest rate guarantee and profit share in Figure 5.6.

Figure 5.6

Rate sensitivity profile of risk policies and bonds, with tangent lines



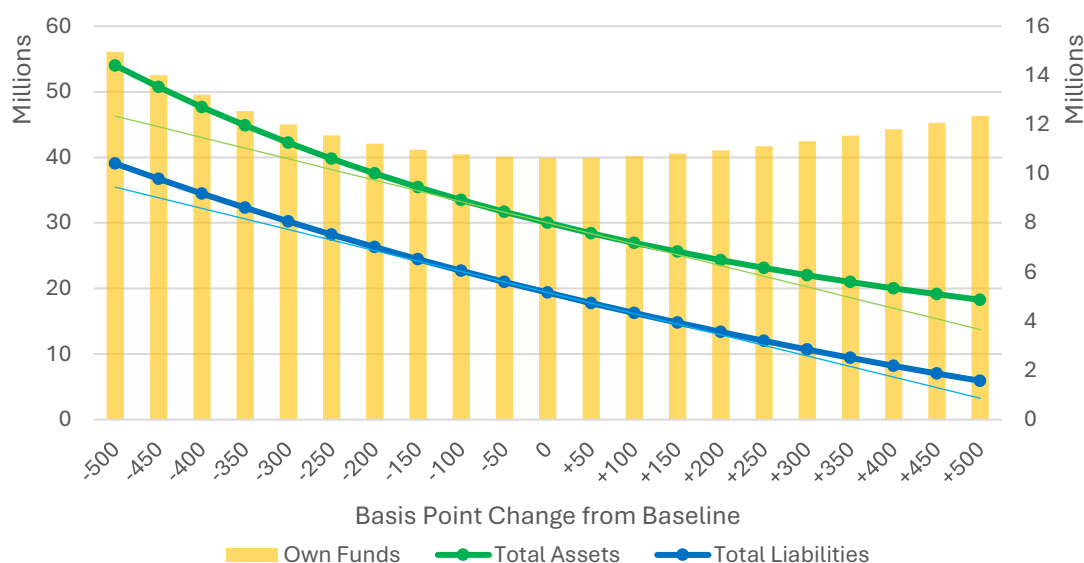
Note. Tangent lines added at zero shift.

The market rate sensitivity profiles are mismatched in Figure 5.6. Notice that if we drew tangents to the blue and green lines at the baseline rate, their slopes, which approximate

negative dollar duration at that point, differ. The large effective duration gaps are shown in Table 5.3. If we disregard disinvestment risk, we can adjust the bond's maturity from 10 years to 16 years, which produces a nearly immunized portfolio shown in Figure 5.7. Unlike participating policies', risk-only products' sensitivity can usually be replicated closely by bonds. Observe Table 5.3 to notice the duration gap fall near zero at the baseline rate after the adjustment.

Figure 5.7

Rate sensitivity profile of risk policies after a duration-matching adjustment



Note. Tangent lines added at zero shift.

Table 5.3

Duration profile for selected bonds, risk policies, and duration gaps, before and after a duration-matching adjustment

Parallel Rate Shift	Assets' Effective Duration before Adjustment	Assets' Effective Duration after Adjustment	Liabilities' Effective Duration	Duration Gap before Adjustment	Duration Gap after Adjustment
-400 bps	8.347	12.318	12.787	-2.303	3.075
-300 bps	8.190	11.953	13.452	-2.484	2.324
-200 bps	8.035	11.586	14.274	-2.661	1.579
-100 bps	7.879	11.217	15.307	-2.835	0.843
0 bps	7.725	10.848	16.629	-3.004	0.119
100 bps	7.572	10.480	18.362	-3.170	-0.590
200 bps	7.419	10.113	20.715	-3.331	-1.280
300 bps	7.267	9.749	24.063	-3.488	-1.947
400 bps	7.117	9.388	29.166	-3.639	-2.590

In their annual financial reports, insurers tend to group risk-only products' reserves with participating policies' reserves, where the former is normally a minor portion. In contrast, the reserves for unit-linked products, which we describe next, tend to be given separately under policyholder-borne risk (Kooprativa pojišťovna, a.s., 2024).

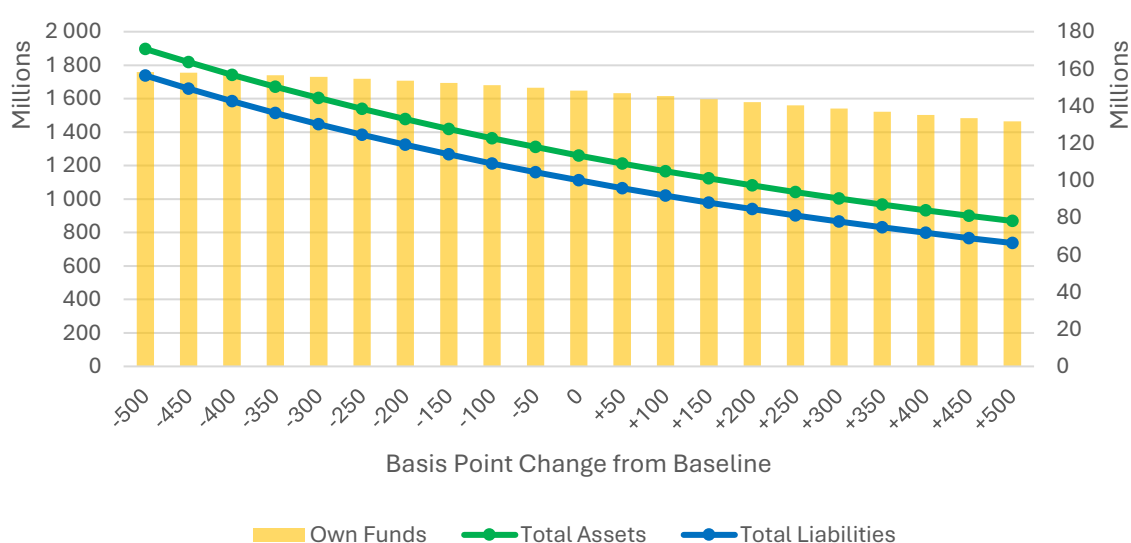
5.3 Life Insurance Unit-linked Products

In unit-linked products, there is no guaranteed interest, and the investment return – contributing fully to the fund value – is determined by the strategy chosen by the policyholder. Unit-linked products fall under *equity-linked insurance* as described by Dickson et al. (2009, p. 374). The name “unit-linked” comes from British terminology where the policyholder is understood as buying units in an underlying asset fund. In case of death while the policy is active, the policyholder's benefit amounts to the sum assured plus their fund value. The benefit paid in case of survival to maturity is their fund value. If the policyholder surrenders their policy, they receive their fund value reduced by a surrender charge.

The policyholder hence bears the risk of poor investment returns. If the underlying funds based on the policyholder's strategy lose value, the insurer's liability decreases correspondingly, as illustrated in Figure 5.8. Differences in behavior between assets and liabilities arise from management fees, the C-2 insurance pricing risk and the C-4 business risk.

Figure 5.8

Rate sensitivity profile of unit-linked products



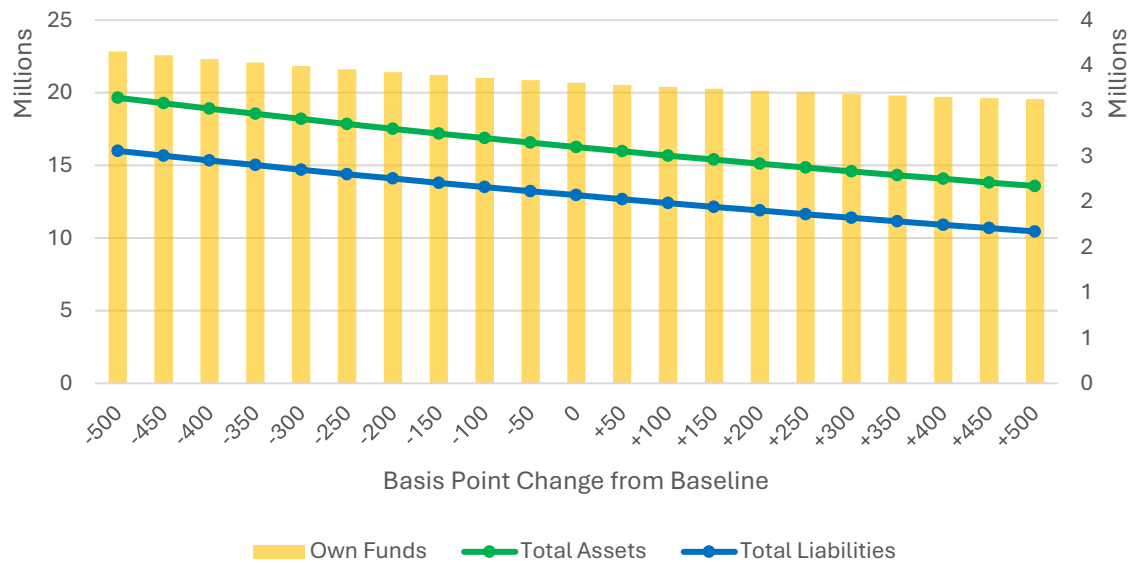
5.4 Non-life products

There is a marked difference in asset management of life and non-life insurance. Life insurers, on average, hold many times more assets than non-life insurers (Cipra, 2006, p. 50). This is because non-life claims are essentially due at the moment of contract's signing – the insured event and the claim timing might easily occur immediately, which makes non-life insurers keep their funds in highly liquid positions with less long-term planning. This is in stark contrast to life insurers and pension funds, who project more predictable cash flows and commit the funds to long-term investments. As we only consider pre-existing business and no future new business, the funds we keep serve to cover the period of time until claim payout.

For Figure 5.9, we assume to start with $N = 50\,000$ policies at a one-time premium of $P = 3\,000$ each. We model all claims to occur and be paid within 10 years. Specifically, $L = 80\%$ (i.e., the loss ratio) of total premiums collected is lost on claims payments, of which $\rho = 70\%$ is paid in the same year as premiums are collected, while $1 - \rho = 30\%$ is carried to the following year as reserve. Of the reserve that enters any following year, ρ is used for claims again, with an exception of the final – tenth year, when all the remaining reserve is used for claims. The first-year cash flow furthermore assumes one-time expenses at $E = 800$ per policy and commissions outflow at $comm = 4\%$ of total premiums received. In summary,

$$CF_t = \begin{cases} N(P - P\rho L - E - P \cdot comm), & t = 1, \\ -\rho(1 - \rho)^{t-1}(LNP), & 2 \leq t \leq 9, \\ -(1 - \rho)^{t-1}(LNP), & t = 10, \end{cases} \quad (5.1)$$

where t is the policy year. We have no guarantees and no profit sharing, so the cash flows are independent of interest rates and we may therefore refer to duration as the discounted mean term. The assets contain 1 820 bonds of face value 10 000 each maturing in 2028; the duration gaps are reasonably low in Table 5.4. The effective duration profile of non-life liabilities as we model them, illustrated in Figure 5.9, is easily replicated with a bond portfolio.

Figure 5.9*Rate sensitivity profile of non-life products***Table 5.4***Duration profile of bonds and non-life products, duration gap*

Parallel Rate Shift	Assets' Effective Duration	Liabilities' Effective Duration	Duration Gap
-400 bps	3.848	4.217	0.428
-300 bps	3.808	4.221	0.397
-200 bps	3.769	4.228	0.367
-100 bps	3.730	4.236	0.338
0 bps	3.692	4.247	0.310
100 bps	3.655	4.260	0.282
200 bps	3.618	4.274	0.255
300 bps	3.582	4.292	0.228
400 bps	3.547	4.311	0.202

Liabilities in non-life products may contain annuities, which would make their interest rate risk profile similar to those in life insurance. We do not model such products in this work.

5.5 Portfolio metrics

Consider a portfolio of 1 063 726 policies such that there are:

- 10 000 profit-sharing policies for the expiry year of 2025 and 5% more for each next expiry year until 2044:

$$N_{Y,PS} = 10\,000 \cdot 1.05^{Y-2025}, Y = 2025, \dots, 2044.$$

The model policyholder for participating policies is a man who entered a 20-year contract at 35 years of age, at the beginning of each of the years 2006, 2007, ..., 2025.

- 1 000 unit-linked policies for the expiry year of 2025 and 5% more for each next expiry year until 2044:

$$N_{Y,UL} = 1\,000 \cdot 1.05^{Y-2025}, Y = 2025, \dots, 2044.$$

The model policyholder for unit-linked policies is a woman who entered a 20-year contract at 40 years of age, at the beginning of each of the years 2006, 2007, ..., 2025.

- 200 000 risk policies, all maturing at the end of the year 2044.

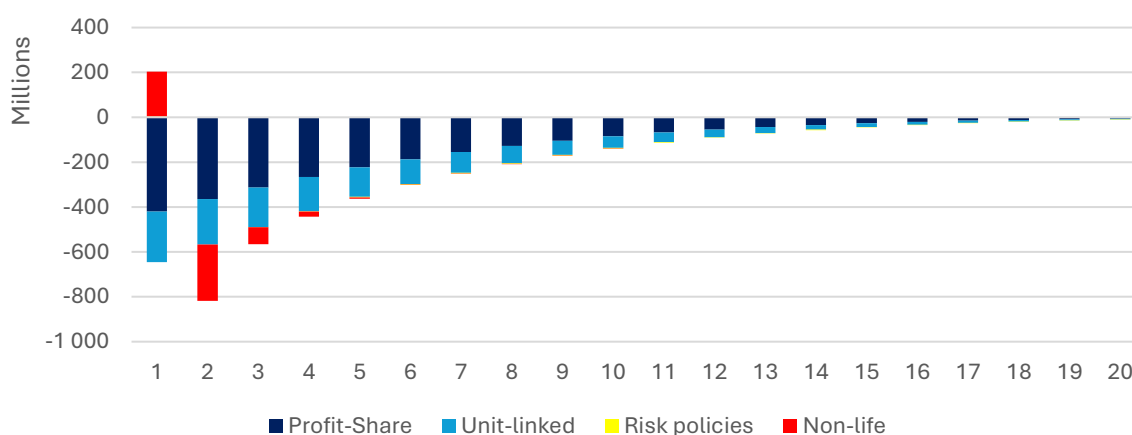
The model policyholder for risk policies, a man in the 5th policy year, was issued the policy when 50 years old at the beginning of 2020 for a period of 25 years.

- 500 000 non-life policies where all claims are paid within 10 years.

For non-life liabilities, model parameters have been retained from Section 5.4, except for the number of policies.

Figure 5.10

Cash flow distribution of liabilities



Note. For a more general presentation, we refer to maturities by years from the present.

For assets, we access the Czech government bonds available on Patria.cz (n.d.). Of the 26 bonds, we exclude three variable-coupon bonds, as well as one with a maturity over 20 years and one bond not denominated in Czech crowns. 21 fixed-coupon bonds remain, the list of which is included in Appendix A.

Let us first attempt cash flow matching. We have already reasoned in Section 4.4.2 that year-on-year matching of asset inflows to liability outflows is unnecessary. Recall equation (4.10); we modify the upper bound by m_b : $m_b[L(b) + \delta_b L(b)] > CF_{Assets}(b) > L(b) - \delta_b L(b) \forall b$. We set $m_b = 1.25$ for all b to permit an equity-to-total-asset ratio of around 20%, consistent with the Czech insurance market as seen in Table 5.6. We will consider six buckets and have flexibility margins defined by δ_b as seen in Table 5.5.

Table 5.5

Liability maturity buckets

Bucket b	Maturity Years	Non-discounted liability	δ_b
1	1	442 652 764	0.05
2	2–3	1 383 102 546	0.1
3	4–5	806 435 408	0.15
4	6–9	931 940 733	0.18
5	10–13	415 334 703	0.23
6	14–20	207 317 819	0.3

Table 5.6

Equity-to-asset ratios of some Czech insurers

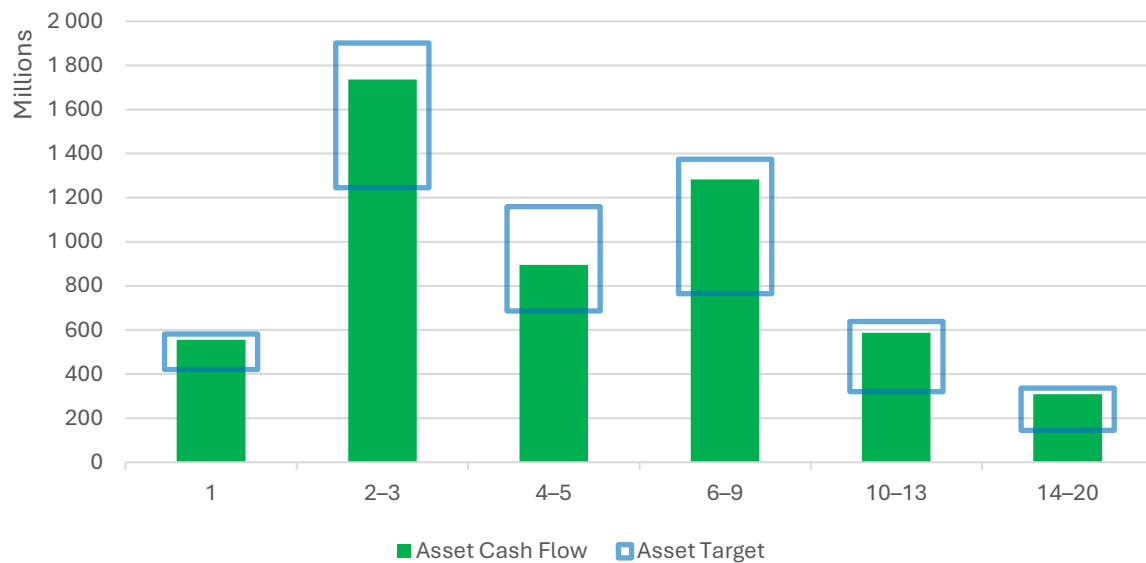
Insurer	Equity to Total Assets Ratio, 2023
Generali Česká Pojišťovna, a.s.	32.4 %
UNIQA pojišťovna, a.s.	18.2 %
ČSOB Pojišťovna, a.s.	18.1 %
Kooperativa pojišťovna, a.s.	17.0 %

Note. Sources: ČSOB Pojišťovna, 2024; Generali Česká pojišťovna a.s., 2024; Kooperativa pojišťovna, a.s., 2024; UNIQA pojišťovna, a.s., 2024.

The asset portfolio, given in Appendix A, will meet cash flow targets for each bucket, as seen in Figure 5.11.

Figure 5.11

Cash flow bucket requirements



Note. The further the maturity from the present, the greater the flexibility range, in our case in relative terms – notice increasing δ_b in Table 5.5. We also aggregate more individual maturities in longer-dated buckets.

In previous sections, we used a constant discount factor and investment return of 5% for the entirety of the previous 10-year horizon. For our 20-year horizon now, we will use the risk-free interest rate term structure for Czechia, dated 31 December 2024, published by the European Insurance and Occupational Pensions Authority (EIOPA, 2025), to calculate discount factors and investment returns. The arising yield curve is shown in black in Figure 5.12, where we also show its parallel shift up and down by 50 basis points in grey. We track the response of assets and liabilities to parallel shifts of magnitude from -350 bps to +350 bps in Figure 5.13.

Figure 5.12

Illustration of parallel 50 bp yield curve shifts

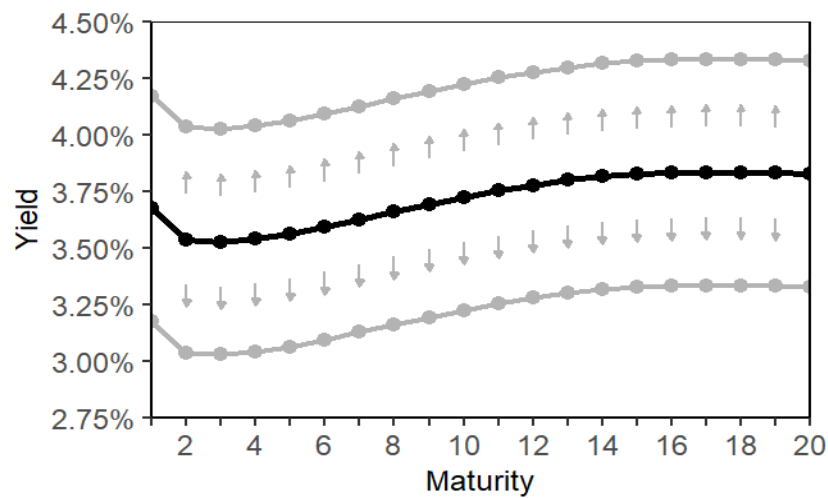
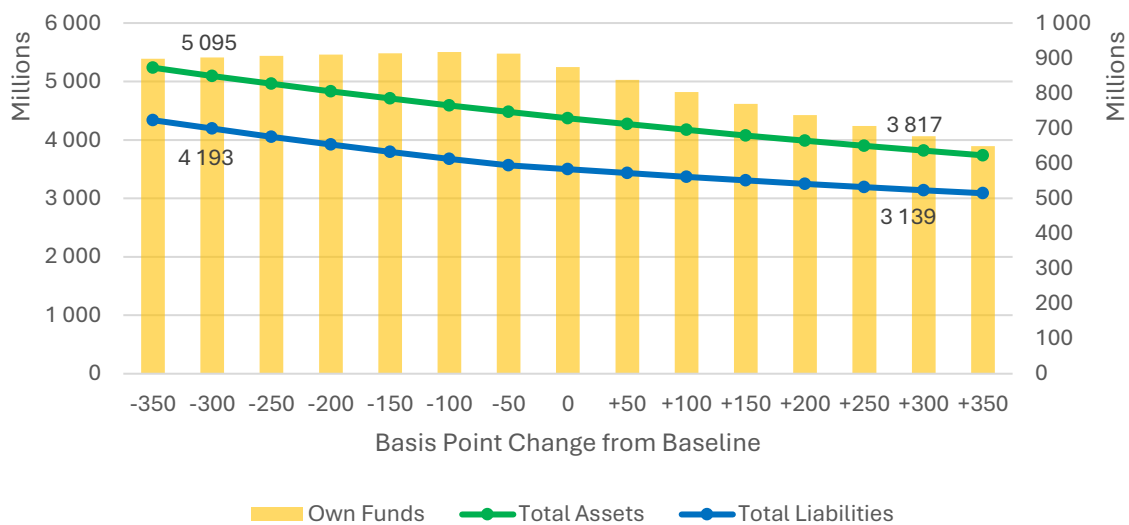


Figure 5.13

Rate sensitivity to parallel yield curve shifts, initial portfolio



Note. Data labels added for assets and liabilities at parallel shifts of -300 bps on the left and +300 bps on the right, identical with the pop-down and pop-up scenarios of NY7 in Table 5.14. Note millions used as units.

5.5.1 Duration Gap

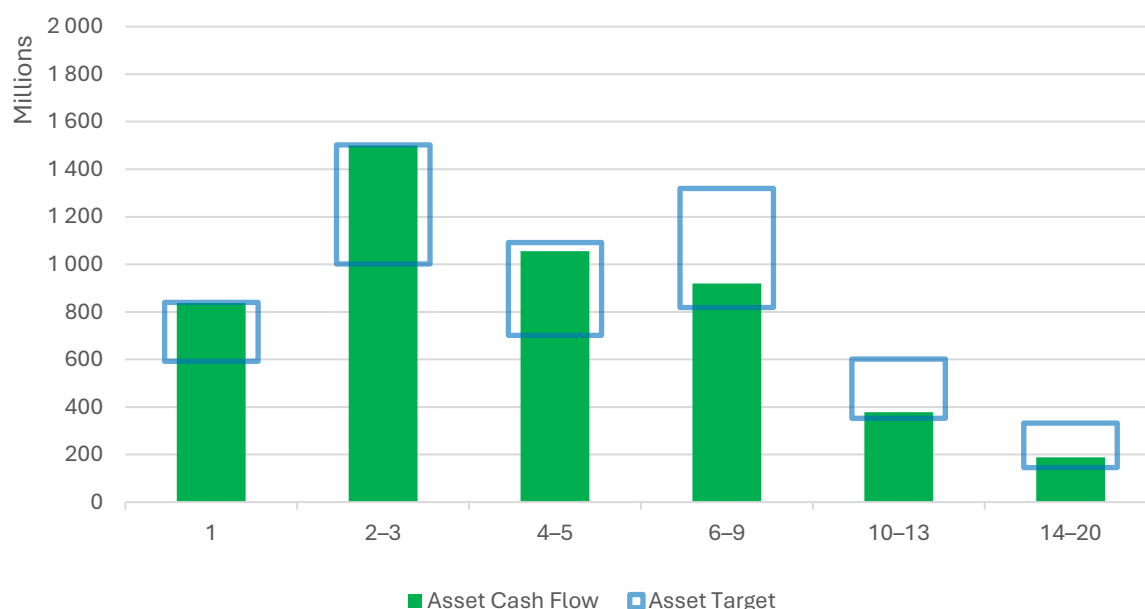
Lengthening asset duration for the portfolio examined in Figure 5.13 by prioritizing long-maturity bonds would protect the equity if we expected rates to fall, but it would produce an

even greater duration gap for positive rate shifts, where the profit share lowers rate sensitivity, which erodes own funds.

On the other hand, shortening asset duration would lower the duration gap at baseline, which is currently 1.686 years. This would lower sensitivity to smaller shifts, such as 10 basis points up or down, which is often desirable. Recalling duration patterns from Section 2.5, we lower the baseline gap by unloading from the long-maturity cash flow buckets in Figure 5.11 in favor of shorter-maturity bonds, and by opting for the greatest-coupon bonds within each cash flow bucket. See an attempt in Figure 5.14 and Figure 5.15, and the change in duration gaps in Table 5.7. We note the changes to bond selection in Appendix A and refer to the shorter-duration portfolio as Portfolio 2, while keeping the name Portfolio 1, also initial portfolio, for the original asset distribution. Note that we preserve the total assets (and keep liabilities untouched), and therefore own funds, at values identical to before the bond redistribution. Specifically, own funds stand at CZK 874 million, which is 20.0% of assets. The market value of liabilities remains CZK 3.498 billion; the market value of assets is CZK 4.372 billion.

Figure 5.14

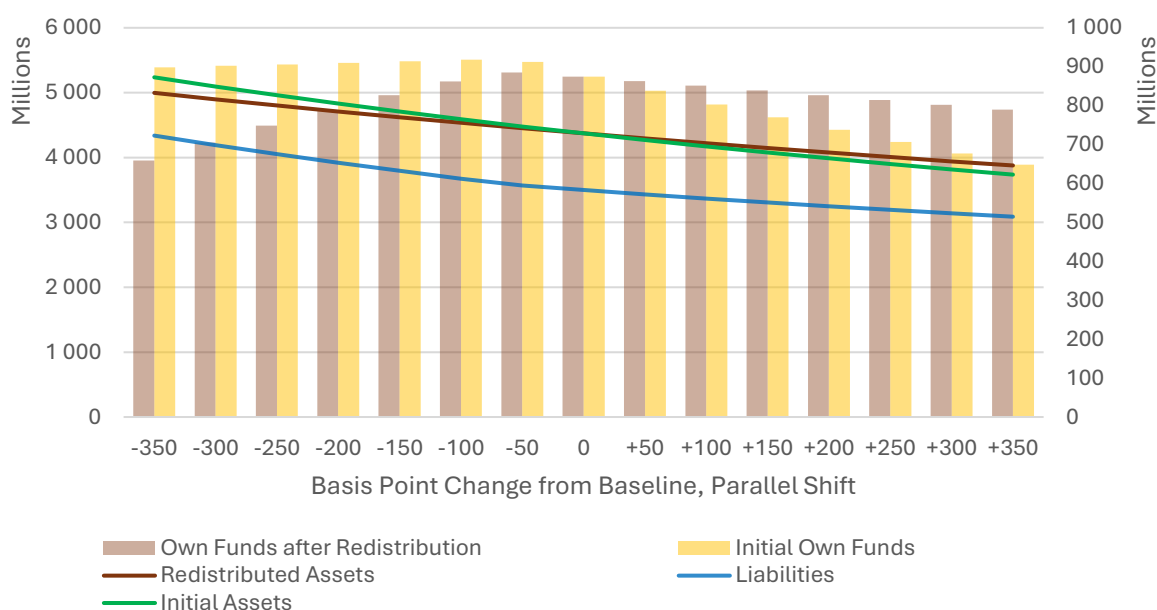
Cash flow bucket matching status after duration-shortening redistribution



Note. Bond redistribution was done in favor of shorter asset duration; shorter maturities, higher coupon rates.

Figure 5.15

Rate sensitivity after duration-shortening redistribution vs. initial portfolio



As seen in Table 5.7, the duration-shortening bond redistribution decreased the duration gap at baseline yield curve by 1.185 years. Notice that equity is comparatively more stable for parallel shifts between -100 and +100 in Figure 5.15 after the adjustment. The adjustment also decreased the duration gap at positive starting yield levels. While the intervention stabilized equity under positive parallel shifts, Figure 5.15 clarifies that the redistribution makes the insurer more vulnerable under negative parallel shifts, such as if the rates returned to near-zero levels seen in mid-2010s.

Table 5.7

Changes in duration profile after shorter-duration redistribution

Parallel Rate Shift	Assets' Effective Duration, Portfolio 1	Assets' Effective Duration, Portfolio 2	Liabilities' Effective Duration	Duration Gap, Portfolio 1	Duration Gap, Portfolio 2
-300 bps	5.415	3.967	6.780	-0.167	-1.855
-200 bps	5.189	3.838	6.611	-0.179	-1.682
-100 bps	4.978	3.718	6.201	0.016	-1.319
0 bps	4.781	3.605	3.869	1.686	0.501
100 bps	4.598	3.498	3.689	1.620	0.545
200 bps	4.426	3.399	3.516	1.562	0.588
300 bps	4.266	3.304	3.351	1.511	0.628

The insurer may set a limit for duration gaps, capping its permissible maximum for all yield curve shifts.

5.5.2 10-point Basis Point Value in Practice

Suppose we achieve a duration gap of zero at baseline rate. If the yield curve moves, assets and liabilities will shift the same way only in relative terms. Absolute changes may be non-zero. Management may prefer absolute changes of assets and liabilities in response to the yield curve moving up or down by 10 basis points – the most readily visualized shift – in a parallel way. As an analogue to dollar duration in Section 2.4, we approximate $\pm 10bpsBPV \approx P(r \pm 0.001) - P(r)$ in Table 5.8. These figures then have a straightforward interpretation of changes in own funds given a parallel yield curve shift of 10 basis points up or down.

Table 5.8

Basis point values of assets, liabilities, and own funds before and after the duration-shortening redistribution

Parallel Yield Curve Shift	Δ Assets, Portfolio 1	Δ Assets, Portfolio 2	Δ Liabilities	Δ Own Funds, Portfolio 1	Δ Own Funds, Portfolio 2
-10 bps	20 987 451	15 836 158	13 586 148	7 401 303	2 250 010
10 bps	-20 804 679	-15 731 240	-13 469 636	-7 335 043	-2 261 604

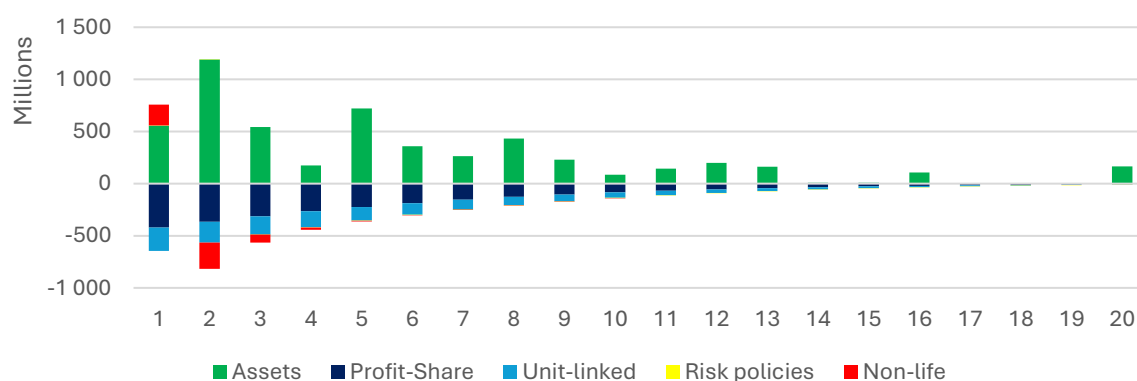
The goal is adjusting the asset portfolio so that the +10bpsBPV and -10bpsBPV of own funds, i.e., Δ Own Funds, are as close to zero as possible – we may impose a maximum on their absolute values. Consistent with narrowing the duration gap somewhat, the duration-shortening bond redistribution described in Section 5.5.1 decreased the +10bpsBPV and absolute -10bpsBPV for equity by over 5 million each, see Table 5.8.

5.5.3 Key Rate Duration Gap

Let us examine the cash flow alignment for Portfolio 1 in Figure 5.16.

Figure 5.16

Cash flow alignment, Portfolio 1



Notice a significant portion of cash flows in the second year, i.e., 2026. If we abandon the assumption of parallel yield curve shifts, the market value becomes a function of several spot rates. We might grow interested in the 2-year spot rate corresponding to 2026. By only moving this rate of the yield curve up and down, we estimate the key rate duration (partial duration) at various interest rates for this single maturity, while all remaining yields remain unchanged. Around the base yield curve as the starting yield level, the key rate duration gap for the 2026 maturity sits at 0.402 years, see Table 5.9. If only the yield at the 2-year maturity shifted up by 100 bps from the current level while the rest of the yield curve remained as it is, as illustrated in Figure 5.17, the value of assets would decrease approximately by 0.493% while that of liabilities would decrease only by 0.115%.

Figure 5.17

Illustration of a 100 bp increase at the 2nd maturity only

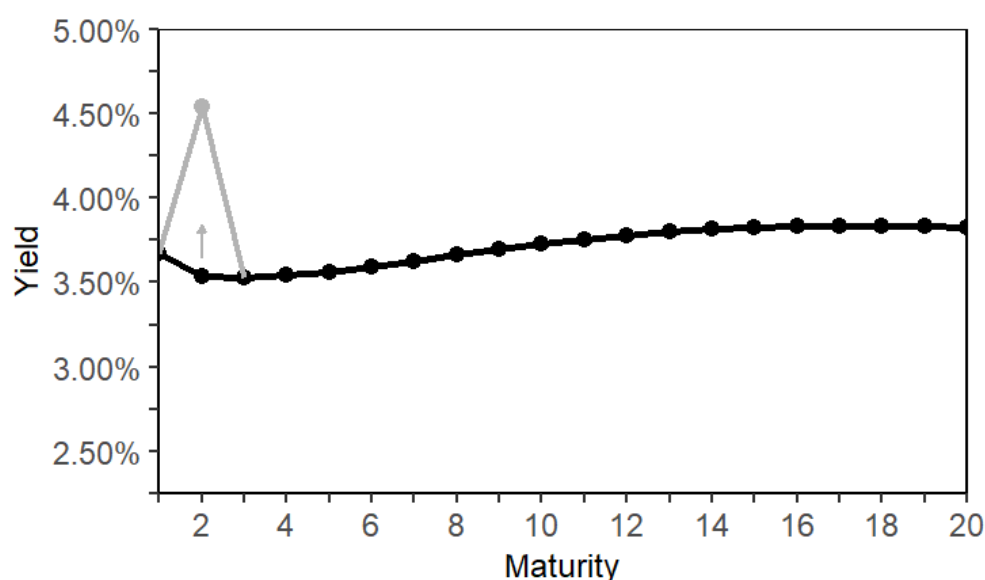


Table 5.9

Key rate duration and gaps for the 2026 maturity, Portfolio 1

Yield Shift	Assets	Liabilities	Key rate duration gap at 2026
-150 bps	0.512	0.561	0.064
-100 bps	0.505	0.538	0.076
-50 bps	0.499	0.318	0.245
0 bps	0.493	0.115	0.402
50 bps	0.488	0.109	0.401
100 bps	0.482	0.103	0.399
150 bps	0.476	0.097	0.398

We also examine which maturities correspond to the largest key rate duration gaps at the starting yield curve level in Table 5.10. The table suggests that the insurer's holdings respond to yield curve changes unevenly, the 2026 maturity being the most sensitive. Recalling Section 2.7, we verify that the sum of key rate durations over all maturities in Table 5.10 indeed equals the effective duration at baseline in Table 5.7 within rounding error. The insurer may set a limit on each year's key rate duration gap to remain below a threshold.

Table 5.10

Key rate durations at baseline yield curve level for each maturity, Portfolio 1

	Key rate year	Assets	Liabilities	Key rate duration gap
1	2025	0.119	-0.334	0.385
2	2026	0.493	0.115	0.402
3	2027	0.326	0.240	0.134
4	2028	0.134	0.347	-0.143
5	2029	0.665	0.429	0.322
6	2030	0.381	0.483	-0.005
7	2031	0.314	0.513	-0.096
8	2032	0.561	0.525	0.141
9	2033	0.319	0.523	-0.100
10	2034	0.126	0.177	-0.015
11	2035	0.221	0.161	0.093
12	2036	0.319	0.142	0.205
13	2037	0.265	0.123	0.167
14	2038	0.013	0.104	-0.071
15	2039	0.013	0.087	-0.056
16	2040	0.189	0.071	0.132
17	2041	0.011	0.057	-0.034
18	2042	0.011	0.045	-0.025
19	2043	0.011	0.035	-0.016
20	2044	0.300	0.026	0.280

We consider the 10-point key rate analogy to the 10-point basis point value. If the yield curve shifted only at the 2026 rate by 10 bps up or by 10 bps down, the assets, liabilities and own funds would change as shown in Table 5.11. These values may be referred to as *key rate sensitivities*: $\pm 10bpsKRS \approx P(r_{2025}, r_{2026} \pm 0.001, r_{2027}, \dots) - P(r_{2025}, r_{2026}, r_{2027}, \dots)$.

Table 5.11*Key rate sensitivities for the 2026 maturity, Portfolio 1*

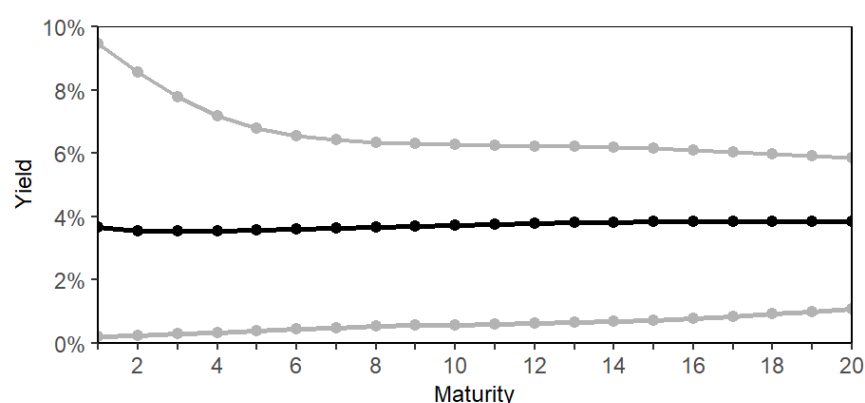
Shift at the 2026 rate	Δ Assets	Δ Liabilities	Δ Own Funds
-10 bps	2 160 431	402 903	1 757 528
10 bps	-2 154 171	-398 618	-1 755 554

5.6 Stress Scenarios

As we have developed the idea of shifting individual points of the yield curve, we can move to standardized whole-curve shifts. With reference to (4.9), we will model equity movements in stress scenarios.

5.6.1 Historical Scenarios

Historical scenarios replicate past stress events. Recall the rate hike discussed in reference to SVB's failure in Section 4.1. In Czechia, the EIOPA-prescribed yield curve increased substantially between July 2021 and July 2022. We additively map the same shock to the 31 December 2024 EIOPA term structure to obtain an inverted yield curve shown in Figure 5.18. To complement by a downward shock of yields returning to the low levels seen following COVID-19 emergency rate cuts, we use the EIOPA curve for Czechia as of 30 June 2020. Applying to Portfolio 1 and Portfolio 2, the balance sheet responses are depicted in Table 5.12.

Figure 5.18*Rate-hike and pandemic shocks on the EIOPA yield curve*

Note. Yield curve shock replicating the rate hike between July 2021 and July 2022 in Czechia is applied to the risk-free EIOPA Czech yield curve dated 31 December 2024. The base curve is in black. Further added is the curve dated 30 June 2020 beneath the base.

Table 5.12*Responses of the portfolios to historical yield curve stresses*

Scenario	Assets, Portfolio 1	Assets, Portfolio 2	Liabilities	Own Funds, Portfolio 1	Own Funds, Portfolio 2
Base Scenario	4 372	4 372	3 498	874	874
Rate Hike	3 783	3 863	3 187	596 (-278)	675 (-199)
Pandemic Shock	5 133	4 935	4 243	890 (+16)	692 (-182)

Note. The 2022 rate hike and the 2020 pandemic shock. The base scenario is identical for both portfolios. Differences from base scenario's own funds are shown in brackets. Millions used as units.

Consistent with findings in Section 5.5.1, own funds in Portfolio 2 respond less to an increase in yields than in Portfolio 1. On the other hand, own funds in the shorter-duration Portfolio 2 decline in the pandemic scenario where they rise in Portfolio 1.

5.6.2 The New York 7 (NY7) Scenarios

A common set of yield curve developments is the New York 7, abbreviated NY7. This set contains the scenarios as described in Table 5.13. We take our 31 December 2024 EIOPA yield curve as base and simulate each scenario in our assets and liabilities models to examine outcomes presented in Table 5.14. Notice that, for Portfolio 1, the pop-up and pop-down scenarios – the 4th and the 7th – are identical to corresponding parallel shifts presented in Figure 5.13. Again, the longer-asset Portfolio 1 fares better in downward shocks (scenarios 5 to 7), while the shorter-asset Portfolio 2 performs better under positive shifts (scenarios 2 to 4).

Table 5.13*Guidelines for the New York 7 (NY7) scenario testing*

	Scenario	Interest rate movement
1	Base scenario	Rates stay level
2	Gradual increase	Rates uniformly increase over 10 years at 50 bps per year, and then stay level
3	Up-down	Rates uniformly increase over 5 years at 100 bps per year, then uniformly decrease over 5 years at 100 bps per year to reach the original rate, and then stay level
4	Pop-up	Instant jump of 300 bps, and then level
5	Gradual decrease	Rates uniformly decrease over 10 years at 50 bps per year, and then stay level
6	Down-up	Rates uniformly decrease over 5 years at 100 bps per year, then uniformly increase over 5 years at 100 bps per year to reach the original rate, and then stay level
7	Pop-down	Instant drop of 300 bps, and then level

Note. Source: Chan & Tse (2018, p. 307).**Table 5.14***Balance sheet outcomes under NY7 scenarios*

	Scenario	Assets, Portfolio 1	Assets, Portfolio 2	Liabilities	Own Funds, Portfolio 1	Own Funds, Portfolio 2
1	Base scenario	4 372	4 372	3 498	874	874
2	Gradual increase	3 858	4 070	3 087	771 (-103)	983 (+109)
3	Up-down	3 989	4 026	3 179	811 (-63)	847 (-27)
4	Pop-up	3 817	3 942	3 139	677 (-197)	802 (-72)
5	Gradual decrease	5 053	4 746	4 085	969 (+95)	661 (-213)
6	Down-up	4 804	4 734	4 005	800 (-74)	729 (-145)
7	Pop-down	5 095	4 898	4 193	902 (+28)	705 (-169)

Note. Differences from base scenario's own funds shown in brackets. Millions used as units.

5.6.3 Solvency II Scenarios

Solvency II legislation gives term structure shifts to be tested for Solvency Capital Requirement calculation in Article 166 and Article 167 of the Commission Delegated Regulation (EU) 2015/35 (2015), where the former prescribes upward shocks at specified maturities, while the latter specifies downward shocks, both shown in Appendix B. The shocks are given in proportional terms. We apply them to the yield curve dated 31 December 2024, illustrating in Figure 5.19, then observe the changes to equity in Table 5.15.

Figure 5.19

Solvency II-prescribed term structure shifts on the EIOPA yield curve

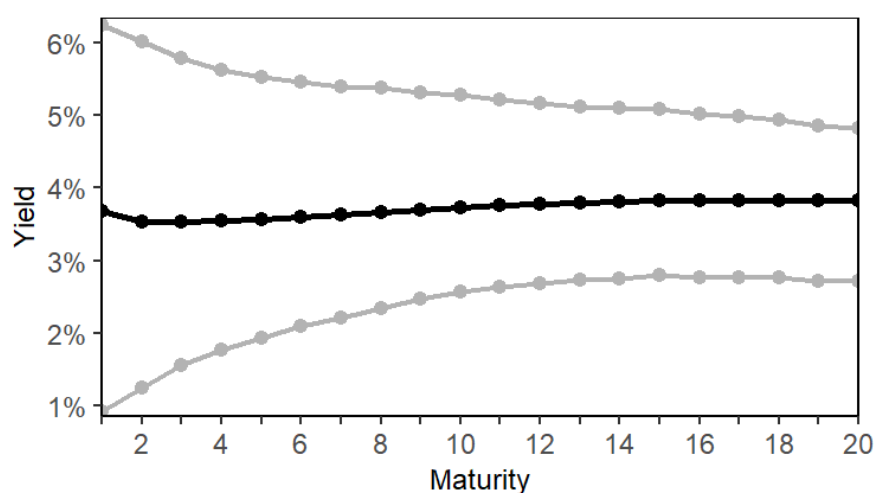


Table 5.15

Balance sheet outcomes under Solvency II-prescribed term structure shifts

Scenario	Assets, Portfolio 1	Assets, Portfolio 2	Liabilities	Own Funds, Portfolio 1	Own Funds, Portfolio 2
Base Scenario	4 372	4 372	3 498	874	874
Upward Shock	4 022	4 077	3 290	733 (-141)	787 (-87)
Downward Shock	4 705	4 652	3 809	896 (+22)	843 (-31)

Note. Millions used as units.

The scenario that results in the larger reduction in own funds – in both portfolio cases, the upward shock – enters the Solvency Capital Requirement calculation.

Conclusion

This thesis reviewed how insurers manage the risk of asset-liability mismatch. We recognized its greatest component to be shifts in interest rates, as Czech insurers place most of their investments in bonds. We set to evaluate how a Czech insurer's balance sheet reacts to interest rate movements with an objective to minimize fluctuations in own funds. We sliced the liability portfolio into 4 types of products. Non-life contracts and risk-only policies were easily hedged by minimizing the duration gap. Unit-linked policies imparted comparatively little ALM risk, because the insurer guaranteed little to nothing on top of the returns achieved by the asset mix policyholders selected. Participating policies presented the greatest interest rate risk.

We blended the policy types into a liability portfolio and proposed requirements for asset cash flows to temper the reinvestment and disinvestment risks. The fitting government bond portfolio was compared to liabilities in terms of selected yield curve sensitivity metrics. To lower the duration gap and the 10-point basis point values, we applied a duration-shortening adjustment on the assets' side. Departing from parallel yield curve shifts, we examined how single-maturity moves drove portfolio changes using the key rate duration. Finally, we presented yield curve stress scenarios in use.

Hedging with the redistribution of tradeable government-issued bonds is insufficient; stabilizing own funds under upward yield curve shifts by redistribution had the opposite effect under downward shocks due to the interest rate guarantee. Tailor-made financial options are too expensive and illiquid in the Czech market. The insurer may alleviate interest rate risk by issuing guarantees which reset every few years, albeit only for new products. Discretion in profit allocation to policyholders is also desirable.

The thesis limited its asset portfolio to the currently tradeable government bonds and assumed their yield to be risk-free. In real life, insurers also hold corporate bonds, which impart credit spread risk, while stocks and real estate come with higher volatility and liquidity risk. Future work could value liabilities and assets with a calibrated interest rate model and Monte Carlo yield curve scenarios to estimate quantile measures for own funds.

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Appendices

Appendix A: Bond Portfolios

The government bond information shown in Table A1 has been retrieved from Patria.cz (n.d.) on 26 April 2025.

Table A1

Selected Czech government bonds

Bond	Maturity year	Coupon rate p.a.	Face value	Yield to maturity p.a.	Number of Bonds in Portfolio 1	Number of Bonds in Portfolio 2
1	2025	2.40%	10 000	3.508%	45 261	36 500
2	2026	1.00%	10 000	3.398%	60 000	0
3	2026	6.00%	10 000	3.322%	50 000	158 000
4	2027	0.25%	10 000	3.400%	48 837	0
5	2028	2.50%	10 000	3.411%	7 000	0
6	2028	5.50%	10 000	3.401%	5 000	97 000
7	2029	0.05%	10 000	3.609%	60 000	0
8	2029	2.75%	10 000	3.546%	5 000	0
9	2029	5.75%	10 000	3.376%	2 000	0
10	2030	5.00%	10 000	3.656%	1 000	79 704
11	2030	0.95%	10 000	3.633%	30 000	0
12	2031	1.20%	10 000	3.728%	22 000	0
13	2032	1.75%	10 000	3.860%	20 000	0
14	2032	4.50%	10 000	3.830%	19 000	0
15	2033	2.00%	10 000	3.985%	20 000	0
16	2034	4.90%	10 000	3.958%	6 000	32 000
17	2035	3.50%	10 000	4.070%	12 000	0
18	2036	4.20%	10 000	4.181%	18 000	0
19	2037	1.95%	10 000	4.293%	15 000	0
20	2040	1.50%	10 000	4.416%	10 000	16 000
21	2044	4.00%	10 000	4.526%	16 000	0

Note. All bonds are denominated in Czech crowns. Yield to maturity is calculated using the sale price.

Appendix B: Solvency II Yield Curve Scenarios

Table B1

Upward and Downward Stress Scenarios for the Basic Risk-Free Interest-Rate Term Structure by Maturity (%)

Maturity (in years)	Increase	Decrease
1	70%	75%
2	70%	65%
3	64%	56%
4	59%	50%
5	55%	46%
6	52%	42%
7	49%	39%
8	47%	36%
9	44%	33%
10	42%	31%
11	39%	30%
12	37%	29%
13	35%	28%
14	34%	28%
15	33%	27%
16	31%	28%
17	30%	28%
18	29%	28%
19	27%	29%
20	26%	29%
90	20%	20%

Note. Source: European Commission (2015).